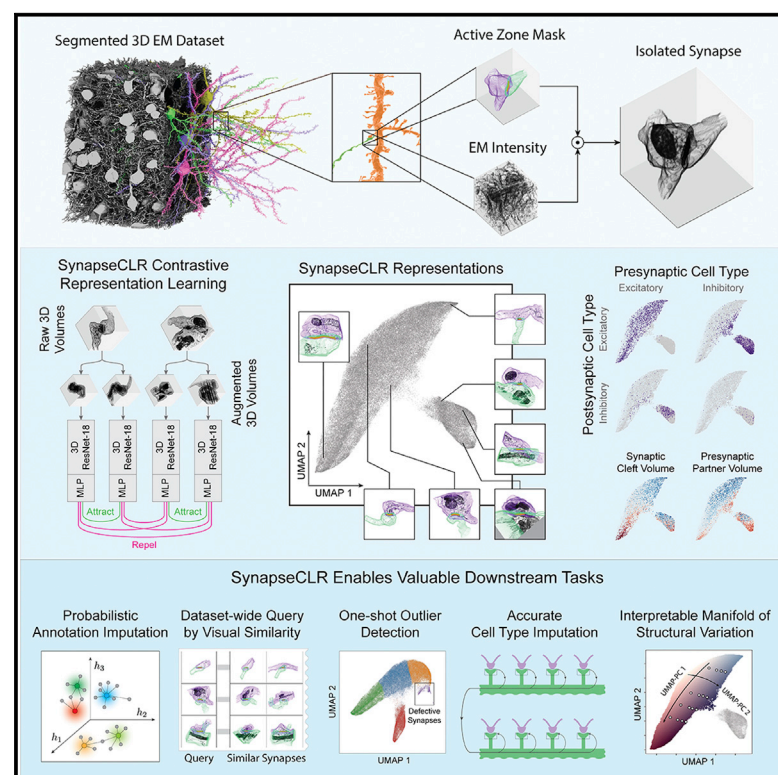


SynapseCLR: Uncovering features of synapses in primary visual cortex through contrastive representation learning

Graphical abstract



Highlights

- SynapseCLR is a generalizable contrastive learning method for segmented 3D EM objects
- SynapseCLR representations encode 3D EM synapse appearance and functional structures
- SynapseCLR enables accurate imputation of synapse properties with only 0.2% labeling
- SynapseCLR enables one-shot identification of defective synapse segmentations

Authors

Alyssa Wilson, Mehrtash Babadi

Correspondence

alyssa.wilson@mssm.edu (A.W.),
mehrtash@broadinstitute.org (M.B.)

In brief

This manuscript presents SynapseCLR, a self-supervised contrastive learning method for extracting information-rich feature representations from segmented 3D EM objects. The method is used to extract features of synapses from a mouse visual cortex dataset; the obtained representations separate synapses by appearance and functionally important structural annotations. SynapseCLR representations streamline valuable downstream tasks, including one-shot identification of defective synapse segmentations, dataset-wide similarity-based querying, and accurate imputation of annotations for unlabeled synapses, using manual annotation of a small fraction of all synapses.



Article

SynapseCLR: Uncovering features of synapses in primary visual cortex through contrastive representation learning

Alyssa Wilson^{1,2,3,5,6,*} and Mehrtash Babadi^{4,5,*}

¹Department of Neurology, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA

²Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA

³Princeton Neuroscience Institute, Princeton University, Princeton, NJ 08540, USA

⁴Data Sciences Platform, Broad Institute, Cambridge, MA 02142, USA

⁵These authors contributed equally

⁶Lead contact

*Correspondence: alyssa.wilson@mssm.edu (A.W.), mehrtash@broadinstitute.org (M.B.)

<https://doi.org/10.1016/j.patter.2023.100693>

THE BIGGER PICTURE This work addresses a major bottleneck in electron microscopy (EM) connectomics, a field aiming to describe complete cellular connectivity patterns in nervous tissue using 3D EM reconstructions of *ex vivo* tissue samples. 3D EM resolves all cells and organelles, informing about nervous tissue composition and function. Insights from 3D EM could aid efforts to identify neurological disease therapies or design better artificial intelligence; however, analysis is currently limited, with often-manual proofreading, exploration, and annotation of individual structures (e.g., synapses) taking years. SynapseCLR aims to automate 3D EM analysis. By producing compact and information-rich vector representations of segmented structures, SynapseCLR accelerates proofreading (by separating defective versus valid segmentations), annotation (with imputation using few manual annotations), and exploration (with dataset-wide querying of segmented objects by structural similarity).



Proof-of-Concept: Data science output has been formulated, implemented, and tested for one domain/problem

SUMMARY

3D electron microscopy (EM) connectomics image volumes are surpassing 1 mm³, providing information-dense, multi-scale visualizations of brain circuitry and necessitating scalable analysis techniques. We present SynapseCLR, a self-supervised contrastive learning method for 3D EM data, and use it to extract features of synapses from mouse visual cortex. SynapseCLR feature representations separate synapses by appearance and functionally important structural annotations. We demonstrate SynapseCLR's utility for valuable downstream tasks, including one-shot identification of defective synapse segmentations, dataset-wide similarity-based querying, and accurate imputation of annotations for unlabeled synapses, using manual annotation of only 0.2% of the dataset's synapses. In particular, excitatory versus inhibitory neuronal types can be assigned with >99.8% accuracy to individual synapses and highly truncated neurites, enabling neurite-enhanced connectomics analysis. Finally, we present a data-driven, unsupervised study of synaptic structural variation on the representation manifold, revealing its intrinsic axes of variation and showing that representations contain inhibitory subtype information.

INTRODUCTION

Digital reconstructions of brain tissue produced using three-dimensional (3D) electron microscopy (EM) are proving to be

rich resources for learning about properties of neural circuitry.¹

3D EM labels the lipid membranes and some proteins in biological tissue, producing digital image volumes in which all cells and organelles and some cytoarchitectural elements—including,



importantly, components of chemical synapses—are resolved (Figure 1A).^{2,3} The combined presence of these features allows for highly contextualized analysis of cellular-level neuronal connectivity, and, for various organisms, this approach has led to new insights into topics such as cell-type-specific nervous system connectivity,^{2–29} development- or learning-mediated changes in neural circuitry,^{30–33} and the fine structure of neuronal and non-neuronal cells.^{34–39}

Although substantial progress has been made in the annotation and analysis of 3D EM data recently, significant bottlenecks to extracting information from these datasets still exist. For perspective, although most existing datasets span physical scales of only 100s of micrometers to 1 mm per dimension (1/1000th of a mouse brain and 1/1,000,000th of a human brain),⁴⁰ they are of such high resolutions (~5–10 nm/pixel [px] per section) that tera- to petabytes of memory are required to store raw images alone.^{3,41} Furthermore, the numbers of biological objects in mid-sized datasets are large: there are typically 10^2 to 10^4 cell bodies, 10^6 cellular pieces (including pieces of neurons, or “neurites,” and pieces of non-neuronal cells), 10^6 to 10^8 of each type of organelle, and 10^6 to 10^8 synapses.² These many objects and the many subtle structural variations among objects of each type⁶ make fully human-based analysis of 3D EM datasets infeasible. For this reason, machine learning approaches, in particular supervised deep learning methods, are becoming indispensable in the 3D EM community. In these approaches, image objects of interest are manually identified and labeled by human experts to generate a training set, which is then used to train specialized deep artificial neural networks to automate the laborious task of object identification and labeling for the rest of the dataset. This framework has enabled generating nearly complete segmentation layers consisting of cells, nuclei, synaptic active zones or “clefts,” or mitochondria for massive-scale 3D EM datasets.^{2,42–45}

To extract neuroscientific insight from segmented and labeled 3D EM datasets, the main approach has been to measure a collection of human-designed features that have previously been identified as structural correlates of function. For example, synaptic cleft size, which is thought to correlate with stronger signal transmission,^{34,46} is often a primary measure used to determine the “strength” of connectivity between neurons. Such features are combined with cell type when that information can be discerned. This is commonly done manually, although some recent efforts have used supervised deep learning to assign presynaptic cell types to synapses,^{47–49} with varying degrees of accuracy. Although this approach has led to insights about neuronal connectivity patterns,^{30,33,39} it is unclear whether other important structural features are being suboptimally accounted for or missed altogether by relying on human interpretation of past work, which is typically based on highly focused labeling or electrophysiological recordings of particular cells. Some structural measures are already known to correlate with one another (e.g., synaptic cleft size and mitochondrial presence),^{50–52} and a description that more comprehensively identifies these features and their quantitative relationships will be an important step toward characterizing neural circuit function.

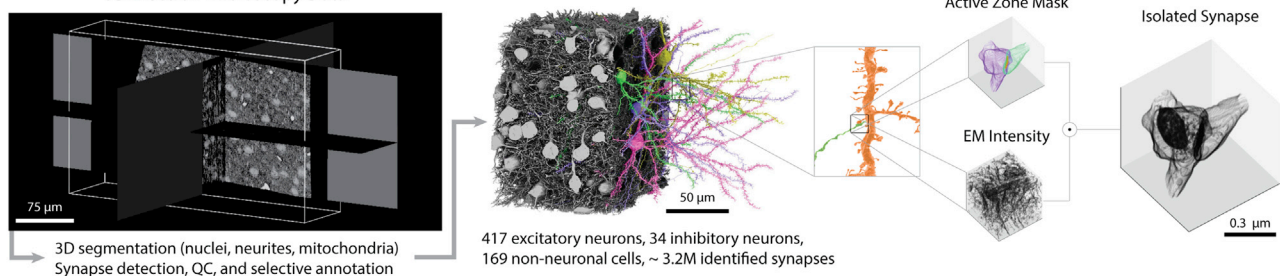
In this work, we take a step toward building a framework for unbiased, data-driven, and scalable analysis of structural neuronal connectivity patterns using recently developed self-su-

pervised learning (SSL) techniques. In contrast to the supervised learning paradigm, where the primary goal is automatic generation of expert-defined image features and annotations for an entire dataset, given a relatively small training set (i.e., learning by example), the goal of SSL is to extract low-dimensional and information-rich image features in the first place. These features, also referred to as representations interchangeably, are obtained via a meta-learning principle such as robustness to noise, or discrimination power (see below). Even though supervised deep learning methods indirectly provide low-dimensional feature representations as a byproduct (e.g., the activation of deeper layers), such representations are largely influenced by the expert-defined features provided as input, rather than the data themselves.

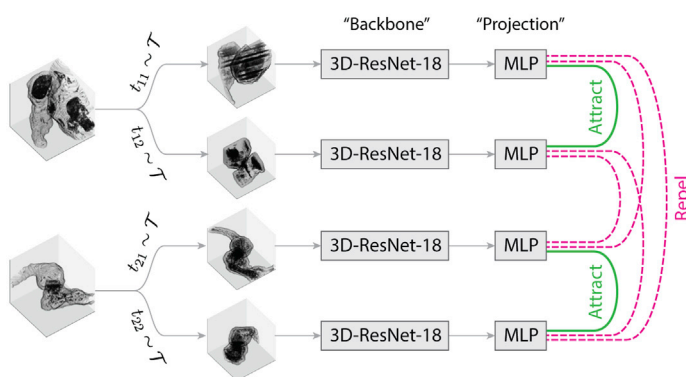
More specifically, we focus on the task of extracting robust and discriminating low-dimensional representations of 3D EM images of chemical synapses (a fundamental component of neural circuitry) without human supervision, and to evaluate the utility of the obtained representations for key tasks such as identification of parent cell types and prediction of previously noted anatomical measurements. Rather than manually identifying presumably important and handcrafted features for downstream tasks and analysis, or using manual labels to drive this feature learning task, we instead define a collection of structural transformations under which synapse identity should remain “invariant.” Our approach follows the SimCLR SSL framework,⁵³ which was originally introduced for learning representations of natural images. For this type of task, we employ a convolutional neural network (CNN), which is a flexible artificial neural network for visual feature extraction. We train our CNN to remove redundant visual features while retaining the most robust discriminating visual features using a mix-and-match pretext task (Figure 1B). Specifically, each input image, in a large batch of randomly selected images, is transformed (“augmented”) twice; all images are then passed forward through the CNN (the “backbone”), followed by an additional shallow multi-layer perceptron (MLP) (the “projector”) for further feature processing, and the resulting representations are used to evaluate and minimize a contrastive loss function. The loss function is designed such that *similarity* between representations from the *same* source image, and *dissimilarity* between representations from *different* source images, both lead to lower loss (see section “experimental procedures”). In practice, SSL frameworks have been shown to produce powerful low-dimensional visual representations that surpass those obtained using other machine learning paradigms, such as supervised learning.⁵⁴ The SSL representations have been shown to be applicable for a wide range of visual tasks, such as classification and segmentation^{55–59} and few-shot transfer learning^{60,61} (i.e., to generalize well to unseen, but related, tasks with just a few examples). The success of the SSL paradigm largely relies on judicious choice of augmentation transformations (e.g., rotation, random cropping, or color histogram distortions), as well as on the implicit “inductive bias” of the employed artificial neural network (i.e., using a network architecture that is specifically designed to extract features of the input data in an efficient way). A natural artificial neural network architecture for processing visual data is the CNN, owing to its desirable translational equivariance and hierarchical feature extraction properties.⁶²

A Dataset Specification and Overview of the Pre-processing Pipeline

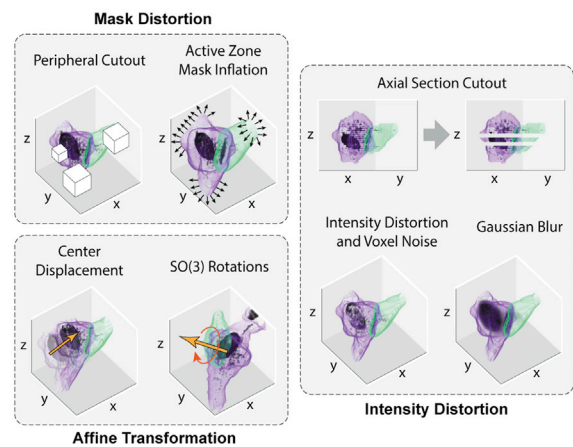
Layer 2/3 Mouse Visual Cortex 250 x 140 x 90 μm
3D Electron Microscopy Data



B Contrastive Representation Learning using SynapseCLR



C SynapseCLR Augmentations



D Application Areas of SynapseCLR Representations

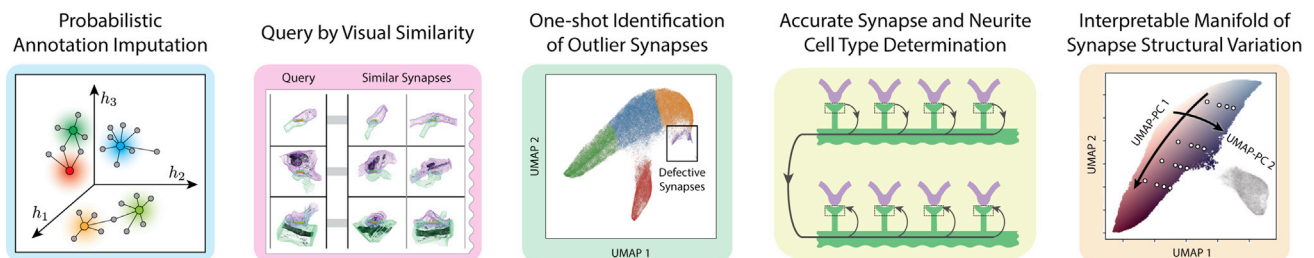


Figure 1. The SynapseCLR workflow

(A) Dataset specification and overview of the preprocessing pipeline. (Left) Cross-sectional images showing the composition of the 3D EM dataset² used to train SynapseCLR. (Middle) A view of this dataset in the same orientation, showing only excitatory neurons (pyramidal cells) with somas in the volume; the left half shows all excitatory somas, and the right half shows four examples in color to illustrate the morphologies of these neurons. (Middle inset) An example of the synapses in these data from which we drew our training image chunks, in this case involving an excitatory neuron dendrite (orange) that is receiving a synapse from an axon (green). (Right) The two raw image chunk layers we used to construct our input image tensors. The top panel shows the segmentation layer, showing only the segmentations of the presynaptic neuron, postsynaptic neuron, and the synaptic active zone (clef) that they form. The bottom panel shows the EM data from the same region. (Far right) We used the segmentation layer to mask the EM, creating an image chunk with the isolated EM image of the synapse only. Our input image tensors were composed of this masked EM chunk. Left, 75 μm ; middle, 50 μm ; right, 0.3 μm .

(B) Contrastive learning workflow. Each input image in the tensor is subjected to two different, stochastically constructed augmentations (see section "experimental procedures"), then passed through a 3D CNN (we used a 3D-ResNet18), and a multi-layer perceptron (MLP) head. The NT-Xent loss (Equation 10 in section "experimental procedures") used to train the network is built such that the difference between outputs for augmentations of the same image is decreased, and that the difference for augmentations of different images is increased.

(C) Augmentations used in SynapseCLR: these include mask distortions, intensity distortions, and affine transformations (see section "experimental procedures").

(D) Application areas of SynapseCLR representations explored in the present paper.

We adapted the SimCLR framework to work on 3D volumetric data rather than two-dimensional (2D) images, along with a set of augmentation transformations specifically designed for

segmented 3D EM data (Figure 1C; section "experimental procedures"). We used the resulting package, called "SynapseCLR," to extract representations from 97,658 synapses

selected from a recently released 3D EM reconstruction of mouse primary visual cortex.² To this end, we used a 3D-ResNet18 CNN^{63,64} as our choice of the backbone network (see Figure S1). This network architecture is the 3D generalization of the ResNet18 CNN⁶⁵ successfully used in earlier work for the purpose of self-supervised representation learning.^{53,54,56,58,59} These representations reveal that synapses are partitioned into several highly distinct regions in the feature space, with smooth variation of representations within each region (see section “[results](#)”). Importantly, by comparing these representations with annotated anatomical measurements for 5,664 of these synapses that were verified previously,² we find that the structures that synapses occupy in feature space correspond to changes in structural correlates of function. Excitatory-to-excitatory synapses, synapses involving inhibitory neurons, and synapses with segmentations contaminated by breaks or other artifacts each occupy distinct regions of the representation space. Within each connected component of the representation space, we observe smooth and correlated changes in synapse size, the presence of pre- and postsynaptic mitochondria, and the distance between a synapse and its postsynaptic neuron. This emergent low-dimensional manifold structure (i.e., organization of synapse representations in such a fashion that they vary smoothly along a few independent directions) allows us to systematically study the structural diversity of chemical synapses, and map out relationships between the various structural annotations we have at our disposal for this study. The strong correlation between these annotations and the representations allows us to use them for a variety of downstream tasks (Figure 1D), including one-shot detection of synapses with defective 3D reconstructions in a dataset, imputation of structural measures for unlabeled synapses by regressing against available annotations, accurate cell type assignment for small neurites that form synapses but otherwise lack the morphological context needed for typing (this application allows those neurites to be included in neural connectivity analysis), and querying of synapses across a dataset based on their visual similarity.

The SynapseCLR package is publicly available on Zenodo: <https://doi.org/10.5281/zenodo.7545417>, and can be readily used for learning representations from other 3D EM datasets. We additionally provide open access to all our trained models, experiments, and feature representations, along with high-quality Jupyter notebooks for reproducing every figure and table presented in this paper (see section “[experimental procedures](#)”).

RESULTS

We pulled synapse images from an EM dataset² consisting of ~2,250 40-nm-thick tissue sections (resolution $3.58 \times 3.58 \text{ nm}^2/\text{px}^2$) stacked into a total volume of $240 \times 140 \times 90 \mu\text{m}^3$ (Figure 1A; section “[experimental procedures](#)”). This dataset has multiple segmentation layers, three of which we used: first, a cellular segmentation containing 451 neurons with somas in the volume and ~8 million neurites; second, a segmentation of ~3.2 million automatically detected synaptic clefts; and third, a segmentation containing ~2 million automatically detected mitochondria. Of the synaptic clefts, 5,664 had been previously verified by experts as representing real synapses, because they were formed between neurons with somas in the volume (398 to-

tal) and were used for previous analysis of connectivity between those cells. Biological annotations had also been made and verified for these synapses, including pre- and postsynaptic cell type (i.e., excitatory versus inhibitory), distance of the synapse from the pre- and postsynaptic somas, synaptic cleft size, and number and size of pre- and postsynaptic mitochondria (i.e., mitochondria located within ~1–5 μm of the synaptic cleft). The distribution and the pairwise correlations of the available annotations are provided in Figure S2 for reference.

To maximize our ability to evaluate and interpret our learned representations, we included all 5,664 of these synapses (called “annotated synapses”) in our training, along with 91,994 additional synapses we randomly selected from the remaining clefts (“unannotated synapses”; see section “[experimental procedures](#)”). These $\sim 10^4$ unannotated examples were located throughout the volume, thus providing a spatially uniform sampling of the synapses across different cell types and different regions within this piece of cortex (layer 2/3 of area V1).

SynapseCLR learns meaningful low-dimensional representations of synapses

We structured each input image beginning from a two-channel 3D image tensor centered at the synaptic cleft centroid. The first channel was an integer encoding of the presynaptic process, synaptic cleft, and postsynaptic process segmentation layers (Figure 1A). The second channel was the volumetric EM image chunk from the same region. Image chunks initially measured 256^3 px^3 at $7.16 \times 7.16 \times 40 \text{ nm}^3/\text{px}^3$ resolution. After augmentation and preprocessing, each image that was ultimately used for training was a 3D image chunk containing only the EM of the presynaptic process, synaptic cleft, and postsynaptic process, and measuring 96^3 px^3 at $14.3^3 \text{ nm}^3/\text{px}^3$ resolution (Figures 1A and 1C; see section “[experimental procedures](#)”).

We masked our EM images to only include pre- and postsynaptic processes because we wanted to learn representations of chemical synapses in isolation. Although the dense neuropil that surrounds each synapse may affect the physiology of that synapse to varying degrees—microglia can modulate synaptic function,⁶⁶ whereas many axons pass closely to synapses without forming any apparent functional interfaces^{4,67}—the entropy of this external material is so high that it would be challenging to learn a meaningful representation of the synapse itself if it were included. Studying potential interactions between synapses and the surrounding processes can be done as a secondary task, which requires having meaningful representations of isolated synapses in the first place (the focus of the present work).

We developed a set of augmentation transformations specific to synapses that included both spatial/geometric and intensity transformations, since combinations of both types lead to the most effective representations⁵³ (Figure 1C; see section “[experimental procedures](#)”). Each augmentation was a stochastic composition of global and per-section brightness adjustment, additive and multiplicative pixel-level noise, Gaussian blur, corner and shell masking cutouts in order to encourage learning part-whole relationships, dilations of the segmentation mask applied to the EM image channel, single-section masking, affine transformations (translations and arbitrary 3D rotations, without shearing or scaling since such deformations would mask known identifying properties of synapses), x axis flip, y axis flip, and x-y

swap. Some of these transformations involve hyperparameters (i.e., parameters that are specified by the user and not optimized or tuned during the main training loop), which were stochastically selected from ranges chosen within reason for each augmentation (see section “[experimental procedures](#)”).

We trained the 3D-ResNet18 backbone and the 2-layer MLP projection head (see [Figure S1A](#)) on all 97,658 synapse image tensors across 4 NVIDIA Ampere A100 graphics processing units (GPUs), each with 96-GB GPU RAM. We used a batch size of 192 and trained the network for 200 epochs over 2 weeks, with a learning rate restart after 100 epochs. The loss function decreased steadily and reached a stable minimum ([Figure S1B](#); see section “[experimental procedures](#)”).

Following successful training, one can take activations from any layer of the SynapseCLR network as a potentially useful representation, with the general expectation that deeper layers contain further processed and aggregated information. Here, we explored the 512-dimensional feature maps from the last layer of the 3D-ResNet18 backbone, as well as the 512-dimensional middle layer, and the 128-dimensional last layer of the projection head. As a first visual study of the manifold structure of these different feature representations, we produced 2D Uniform Manifold Approximation and Projection (UMAP)⁶⁸ embeddings at each of these three layers. In brief, UMAP is a general manifold learning technique that produces approximate low-dimensional layouts (embeddings) of high-dimensional data while preserving the global connectivity structure of the given set of data points (here, the representations). We found the UMAP embeddings at all three layers to be structurally consistent ([Figure S3](#), rows 1, 4, and 5). A more detailed study of the global structure of SynapseCLR representations in relation to the available structural annotations is given in later sections. Quantitative benchmarks revealed that the 512-dimensional backbone features provided the highest utility for predicting biological properties of unannotated synapses ([Table S1](#); to be discussed later). As such, we base our main discussion and downstream analyses on 3D-ResNet18 backbone last-layer features; unless explicitly specified, we hereafter refer to this layer when we mention SynapseCLR representations. We briefly comment on the potential utility of other features later (section “[discussion](#)”). We explored both 2D ([Figure 2G](#)) and 3D ([Figure S4](#)) UMAP embeddings of these representations and found, in both cases, these representations occupied several highly separable regions in feature space. A detailed analysis of synapses in each region is provided later; examples from each region are shown in [Figures 2B](#) and [2C](#). In order to assess the robustness and reproducibility of SynapseCLR representations, we repeated the training run from scratch with a different random initialization of both the 3D-ResNet18 backbone and the projector network. In addition, as a basic check of sensitivity to augmentation transformations, we changed one augmentation: namely, we increased the probability of sectional cutout augmentations ([Figure 1C](#)) from 0.5 to 1.0 in the repeated training run (see section “[experimental procedures](#)”), such that both views of any given synapse would have several z sections censored at random. The UMAP embedding of representations obtained from this second training run is shown in [Figure S3](#) (second row) along with colorizations corresponding to several structural annotations. We find the embeddings to be strikingly similar to the first training run and quan-

titative assessments of downstream task performance to be statistically indistinguishable (see below for details).

As a first visual assessment for the meaningfulness of the learned representations, we selected a few typical synapses from the dataset as “prompts,” and “queried” similar synapses based on Euclidean distance in the representation space ([Figures 2D–2F](#) and [S5](#)). The locations of the prompt synapses and their top-seven similarity hits are also shown on the UMAP embedding ([Figure 2G](#)). We noticed that the nearest neighbors shared staggeringly similar structural features to the prompt synapses, providing a first demonstration that SynapseCLR representations highlighted features that capture structural variability across synapses in the dataset.

SynapseCLR representations allow one-shot identification of contaminated synapse segmentations

In our inspection of regions in these UMAP projections, we discovered that one in particular contained primarily synapses with segmentations that were contaminated by artificial breaks or other imperfections ([Figure 2C](#)). By performing Leiden⁶⁹ clustering directly on the 512-dimensional representation space (as shown in [Figure 2A](#)), we found that several types of contaminated synapses formed a single cluster (the purple-colored cluster in [Figure 2A](#)), allowing us to remove them from the dataset in a single operation (“one shot”) along with one nearest neighbor as a conservative quality control (QC) measure. The number of annotated and unannotated synapses passing QC amounted to 5,623 and 89,251, respectively. The removed contamination amounted to 0.7% of proofread and manually annotated synapses, compared with 3% of the remaining synapses, demonstrating the specificity of this QC measure. We did not find synapses exhibiting similar issues in other regions of the representation space.

SynapseCLR representations correlate with meaningful anatomical features

To assess the relationship between these learned representations and anatomical measures that are conventionally used as structural correlates of function, we overlaid each property measured for the 5,623 high-quality annotated synapses ([Figure 2G](#)) on our 2D UMAP embedding ([Figures 2H](#) and [2I](#)).

We found that synapses formed between different neuron types (excitatory [E] or inhibitory [I]; see section “[experimental procedures](#)”) occupy largely distinct regions of representation space ([Figure 2I](#)). Specifically, E-to-E synapses dominate the larger, upper manifold component, and synapses involving inhibitory neurons (I to E, E to I, and I to I) dominate the lower manifold component as well as much of the thin bridge connecting the two. Among the domains inhabited by synapses involving inhibitory neurons, we observe some additional type-specific representation partitioning: I-to-E synapses appear to concentrate in the lower part of the lower component, whereas E-to-I and I-to-I synapses appear to be mostly excluded from this sub-region. This representational organization implies that synapses have inherently distinct structures depending upon whether they are formed by an excitatory or inhibitory neuron and upon whether the synaptic target is an excitatory or inhibitory neuron, consistent with previous structural neuroanatomy findings (see section “[discussion](#)”).

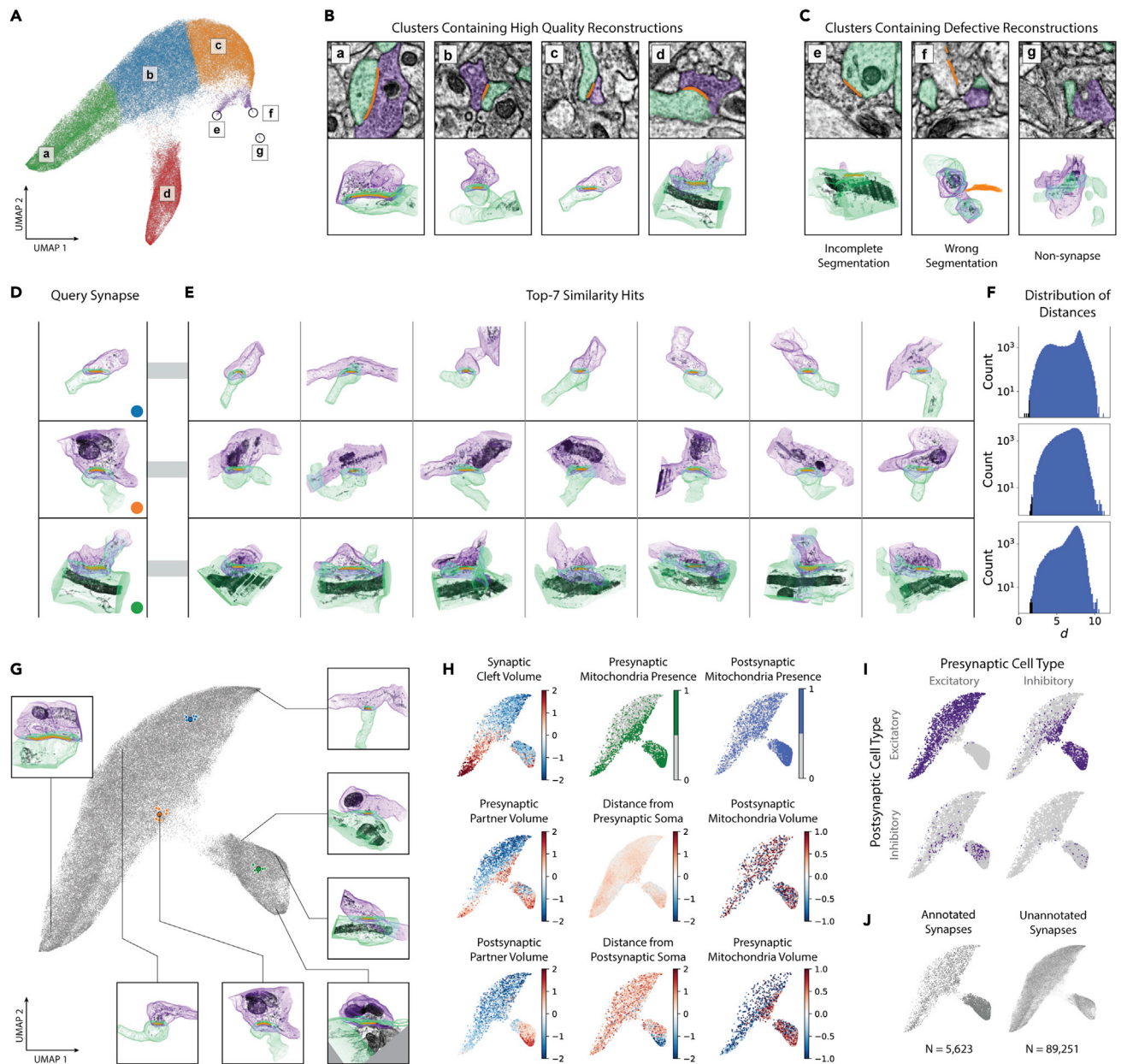


Figure 2. Relationships between SynapseCLR representations and structural properties

(A) 2D UMAP embedding of representations obtained from the trained 3D-ResNet18 backbone features, clustered by building a k-nearest-neighbor graph (k = 100) according to cosine similarity followed by Leiden community detection. Circles around e, f, and g are guides for the eye. The colors correspond to different Leiden clusters.

(B) Example images of synapses from clusters that contain primarily high-quality reconstructions (a–d).

(C) Example images of synapses from clusters that contain defective reconstructions (e–g).

(D) Segmentation-masked EM for three randomly selected synapses used as prompts to query structurally similar synapses in the representation space. These synapses and their nearest neighbors are highlighted in (G) according to the color indicated on the lower-right corner of the query synapse.

(E) The seven nearest neighbors in representation space for each prompt synapse in (D), showing remarkable visual similarities to the corresponding prompt synapse.

(F) Representational Euclidean distances between each synapse and the corresponding prompt synapse shown in (D). Distances for the seven nearest neighbors shown in (E) are colored black in the histogram.

(G) 2D UMAP of representations for only non-defective synapses (94,874 in total). The highlighted group of points correspond to the prompt synapse (outlined points) and the top seven nearest neighbors shown in (D).

(H) Each structural property measured for annotated synapses, superimposed on the 2D UMAP.

(I) Points representing the locations of annotated synapses formed between neurons of various pre- and postsynaptic cell types (dark purple) in the UMAP, showing substantial separation of synapses in representation space as a function of pre- and postsynaptic neuron type.

(J) The density of the annotated and unannotated synapses in the SynapseCLR representation space.

Within each of the two largely separate representation manifold components, we also observed continuous variation of other annotated measures. Notably, synaptic cleft size varied smoothly along the longest dimension of the upper (primarily E-to-E) component. We observed a similar but less dramatic change in cleft size in the primarily I component, again along the longest dimension. The sizes of presynaptic and postsynaptic partners also co-varied somewhat with cleft size in both components (Figure 2H, left column), but, notably, we see that large presynaptic and postsynaptic interfaces do not always correspond with a large cleft size (e.g., at the bottom of the lower component).

We also observed orchestrated variation in whether mitochondria were present in pre- and postsynaptic partners, and in the volume of those mitochondria that was captured in our image chunks (which is a rough proxy measure of how close mitochondria are to the synaptic cleft). In particular, presynaptic mitochondria generally co-occurred with large presynaptic processes in synapses of any type (Figure 2H top middle, left middle).

In contrast, the learned synapse representations captured limited information related to synaptic distance from the presynaptic soma (Figure 2H, middle). Distance from the postsynaptic soma also varied little except at the bottom of the I component, where distances were minimal compared with those for other synapses (Figure 2H, bottom middle).

To further ascertain the value of our SynapseCLR representation learning strategy, we also obtained baseline representations using two other CNNs: a random (untrained) 3D-ResNet18 CNN, and MedicalNet,⁷⁰ a general-purpose pretrained 3D-ResNet18 CNN for 3D segmentation of MRI and computed tomography (CT) datasets in any of a diverse set of organs (including brain). Our inclusion of MedicalNet as a baseline is motivated by the success of transfer learning approaches, i.e., the empirical observation that CNNs trained on images from one domain, e.g., the ImageNet natural image dataset,⁷¹ exhibit strong generalization to other imaging data domains, such as histopathology slides⁷² and cell morphology images.⁷³

The resulting UMAP embeddings of SynapseCLR representations and the two baselines, along with several overlaid annotations, are given in Figure S2. We notice significantly stronger correlation and structure in SynapseCLR representations compared with the two baselines. A quantitative benchmark of these representations for imputing anatomical annotations is presented in the next section.

Given that our SynapseCLR representations are robust descriptors of variation among synapses in the dataset (by design), we conclude that annotations that vary smoothly across representation space (such as synapse size and mitochondrial presence) are indeed important factors in establishing the structural identity of a synapse.

SynapseCLR representations accurately impute properties of unannotated synapses

The strong correlations we observe between SynapseCLR features and anatomical annotations suggest the possibility of using these features to predict (impute) annotations for unannotated synapses (which comprise the vast majority of synapses in our dataset). Endowing synapses with anatomical data is useful for bypassing the need for manual annotation and for increasing

the number of synapses that can be used to assess connectivity in the 3D EM dataset.

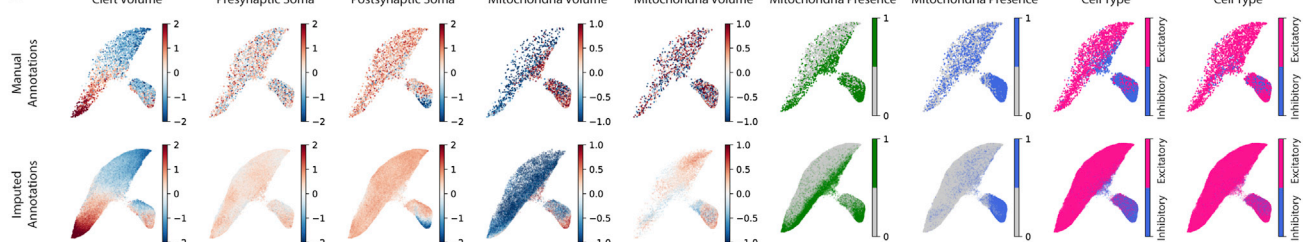
We note that our set of annotated synapses is a biased sample of all synapses in this cortical volume. More specifically, the annotated set is composed of only synapses between neurons with somas in the EM volume, i.e., synapses between local neurons. Neighboring excitatory neurons and neighboring inhibitory neurons are less likely to form synapses with each other than with farther-away cells^{2,41}; therefore, the annotated set is disproportionately enriched with I-to-E and E-to-I synapses. This sampling bias can also be seen in the SynapseCLR representation space by noting the densities of annotated versus unannotated synapses (Figure 2J) in the locations of I-to-E and E-to-I synapses (Figure 2I). Despite this global sampling bias, annotated and unannotated synapses show rather uniform distributional mixing in the SynapseCLR representation space (see Figure 2J). Quantitatively, we find that, for every unannotated synapse, the nearest neighbors in representation space that occupy 5% of the surrounding manifold compartment volume consist of at least 10% annotated synapses. Thus, although other synapses in the volume may be formed by physically more distal cells of differing subtypes, there are no systematic structural features they possess that are not also represented in the annotated synapse population. For this reason, annotated synapses can be used to accurately impute properties for these other synapses.

To generate imputed values of each type of annotation for our unannotated synapses, we trained Gaussian process (GP) regression or classification models using the SynapseCLR representations as covariates and the annotations from our subset of 5,623 annotated synapses as targets. We optimally regularized each model, including the choice of GP kernel, using a cross-validation strategy. In order to correct for the discussed global sampling bias of the annotation set, we balanced the training set of categorical annotations (i.e., pre- and postsynaptic cell types and mitochondria presence) by sub-sampling the same number of synapses from each class during training (see section “experimental procedures”). For comparison, we also trained regression models using representations from the two other baseline CNNs. We found that the backbone-level representations from the trained SynapseCLR CNN generated the best predictions overall (Figure 3A; Table S1; section “experimental procedures”). In the best models, the area under the receiver operating characteristic (ROC) curves (AUC) for predicting pre- and postsynaptic cell types were 0.975 ± 0.004 and 0.940 ± 0.004 AUC, respectively (section “experimental procedures”). The presence of pre- and postsynaptic mitochondria could also be predicted with high accuracy (0.939 ± 0.015 and 0.857 ± 0.001 AUC, respectively). Another annotation that could be predicted well was synaptic cleft size ($70.0\% \pm 2.7\%$ explained variance). On the other hand, our models for a synapse’s distance from its pre- and postsynaptic neuron somas explained only $15.9\% \pm 1.4\%$ and $53.9\% \pm 1.6\%$ of the variance in those measures, respectively. Looking at the scatter plots of predicted versus actual pre- and postsynaptic soma distances (Figure 3C), it seems that both regression models are similarly limited in their ability to impute distances except for the subset of synapses that have small distances to the postsynaptic soma (the blue regions in the third column of Figure 3B). Overall, the diminished predictability of synaptic distances to somas from their structure

A

Method	Synaptic Cleft Volume (EV)	Distance from Presynaptic Soma (EV)	Distance from Postsynaptic Soma (EV)	Presynaptic Mitochondria Volume (EV)	Postsynaptic Mitochondria Volume (EV)	Presynaptic Cell Type (AUC)	Postsynaptic Cell Type (AUC)	Presynaptic Mitochondria Presence (AUC)	Postsynaptic Mitochondria Presence (AUC)
Random 3D ResNet-18	40.887 ± 1.954	2.028 ± 0.309	46.232 ± 1.867	21.872 ± 4.945	1.013 ± 1.285	0.858 ± 0.015	0.778 ± 0.015	0.813 ± 0.025	0.813 ± 0.008
Pretrained [MedicalNet]	49.204 ± 2.868	9.297 ± 1.683	53.262 ± 1.225	41.914 ± 4.548	5.080 ± 1.782	0.917 ± 0.027	0.836 ± 0.016	0.852 ± 0.020	0.844 ± 0.015
SynapseCLR [Ours]	70.017 ± 2.669	15.938 ± 1.426	53.902 ± 1.584	56.869 ± 3.091	9.968 ± 2.402	0.975 ± 0.004	0.940 ± 0.004	0.939 ± 0.015	0.857 ± 0.010

B



C

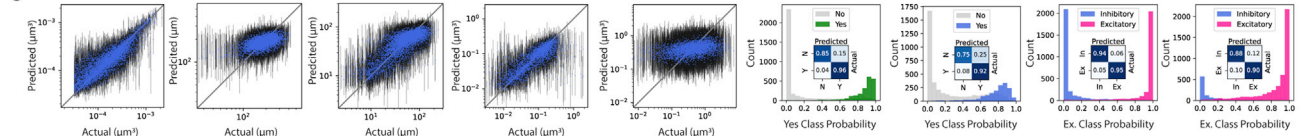


Figure 3. Predicting structural properties of unannotated synapses using SynapseCLR representations as covariates and GP regression

(A) Annotation prediction accuracy for the best GP regression models trained on synapse representations using (top row) a random 3D-ResNet18, (middle row) a general-purpose pretrained 3D-ResNet18 CNN for 3D segmentation of MRI and CT datasets (MedicalNet), and (bottom row) SynapseCLR 3D-ResNet18 backbone representations. Light gray columns, explained variance (EV) for continuous measures; dark gray, area under the receiver operating characteristic curve (AUC) for categorical measures; bold, the best-accuracy model for each annotation.

(B) UMAP embedding of SynapseCLR 3D-ResNet18 backbone representations with structural properties on the subset of annotated synapses (top), and imputed structural properties for all synapses (bottom).

(C) Imputed versus measured structural values for annotated synapses. Blue markers show the GP posterior mean and the black error bars show the 90% credible intervals. The rightmost four panels correspond to binary measures. The histograms show the predicted probabilities stratified by true class labels; the inset plot shows the binary classification confusion matrix.

suggests that the correlation between these two features is subtle. Our models for predicting pre- and postsynaptic mitochondria volumes explained $56.9\% \pm 3.1\%$ and $10.0\% \pm 2.4\%$ of the variance in those measures, respectively. The limited predictive power of these models likely stems from our use of small synaptic image chunks. We discuss potential future improvements in the section “discussion” section.

To further probe the reproducibility of imputation accuracies across different SynapseCLR training runs, we performed the same imputation experiments on the representations obtained from our separate SynapseCLR training run with a different random neural network initialization and stronger augmentations (as discussed earlier). The imputation cross-validation accuracies for both runs are given in Table S2 and are statistically indistinguishable.

Previously, we noted the distributional mixing of annotated and unannotated synapses within each component of the SynapseCLR representation space, despite the global sampling bias in the annotated synapse set at the level of pre- and postsynaptic cell types. We hypothesized that a SynapseCLR model trained exclusively on the unannotated synapses could be used to represent the features of the annotated synapses just as well as a model trained on both annotated and unannotated synapses, and therefore would result in negligible loss in imputation accuracy. To test this hypothesis, we repeated the SynapseCLR training run from scratch only using the unannotated synapses, used the trained model to extract feature representations for

held-out annotated synapses, and repeated the above imputation experiments. The UMAP embedding from this experiment is shown in Figure S2 (third row) and is highly similar to the embedding obtained using all synapses (first row), both in global and local structure. Imputation accuracies are shown in Table S2 along with standard deviations obtained by cross-validation. A one-sided z test shows that the only two annotations exhibiting statistically significant drops at the significance level of 0.05 are pre- and postsynaptic cell types ($p < 6.0 \times 10^{-4}$ and $p < 3.0 \times 10^{-7}$, respectively). Nevertheless, the mean drops in these accuracies are small, at 1.3% and 2.1% respectively. In summary, this experiment demonstrates that the contrastive structural features of unannotated and annotated synapses are indeed highly similar, and that the accuracy of imputation models is not due to “fine-tuning” of SynapseCLR features to the annotated synapses.

Consensus calls of imputed cell types from multiple synapses increases the population of analyzable neurites in 3D EM datasets

Of these imputed annotation values, the ones that are most important for growing a connectomics dataset are the pre- and postsynaptic cell types: E and I neurons are generally understood to have very different functional roles in neuronal circuitry, so any neurite that cannot be typed is excluded from analysis because its role in the circuit cannot be ascertained. However, this conservative approach leads to a severely reduced picture of neuronal circuitry. Many of the synapses in this EM dataset

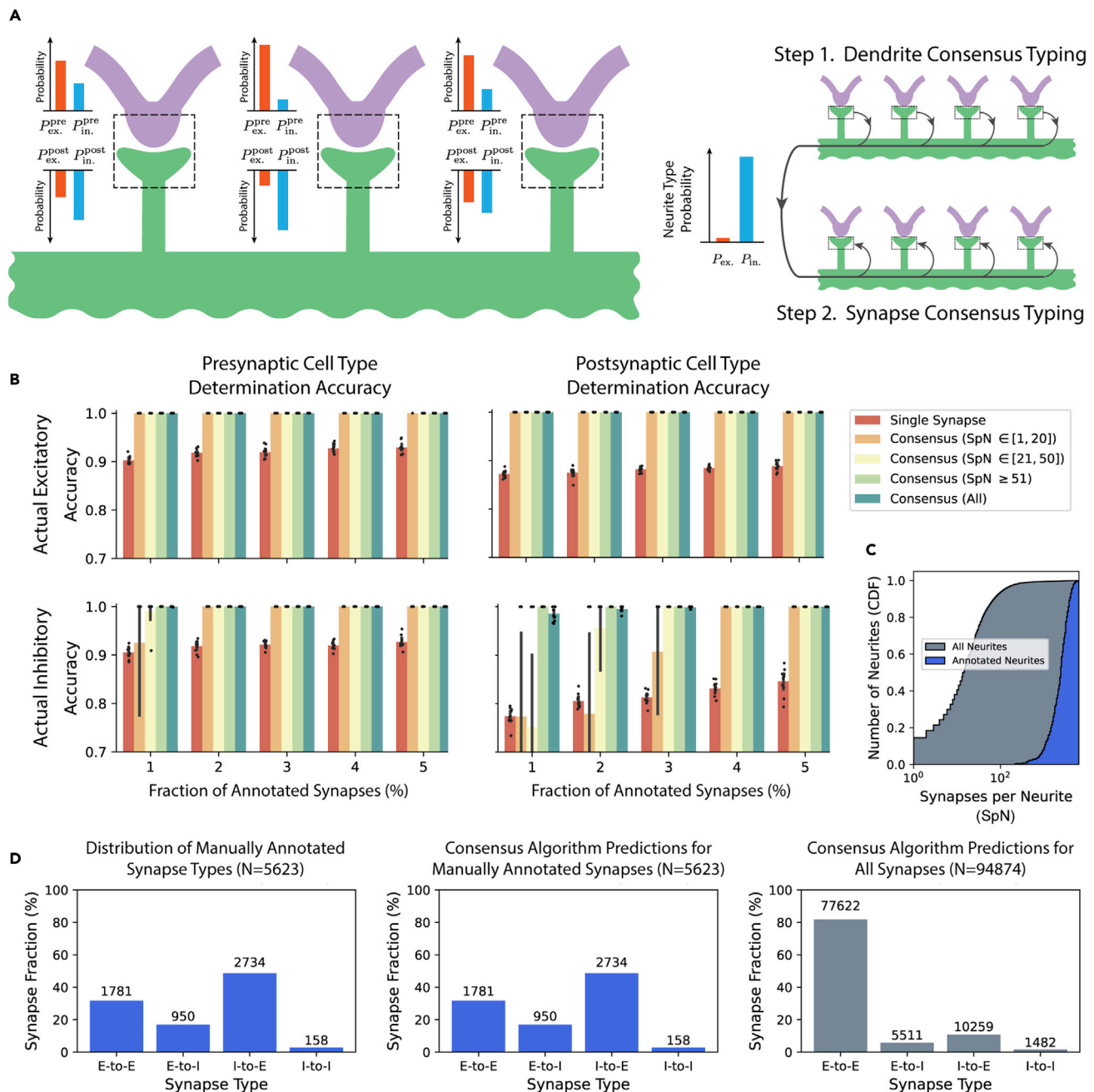


Figure 4. Accurate, consensus-based pre- and postsynaptic cell type determination at single-synapse resolution

(A) (Left) A schematic illustration of how pre- and postsynaptic cell type probabilities, inferred from single-synapse EM data, can exist for several synapses that involve a single neurite (here, formed onto the same dendrite). (Right) The two-step consensus cell-typing algorithm. First, we determine the cell type of each neurite with high accuracy by treating each synapse as a weak classification and ensembling them; next, we propagate the neurite cell type back to the participating synapses (see section “experimental procedures”).

(B) Consensus-based synapse cell-typing cross-validation accuracy, stratified by neurite polarity (pre- or postsynaptic) and actual neuron type (shown in four separate panels); the number of synapses per neurite (SpN) (shown as differently colored bars); and for different fractions of manually annotated synapses in the dataset, i.e., training data size (x axis values). The error bars indicate the inter-quartile range of each accuracy, as obtained by 10-fold train/test partitioning of the annotated synapses. Individual cross-validation runs are shown as dots. Note the dramatic increase in accuracy of consensus calls overall (orange, yellow, and green bars) compared with single-synapse cell type calls (red bars).

(C) The cumulative distributions of SpN for all neurites (gray) and for the neurites associated with annotated synapses (blue) in our dataset. The latter group of neurites are distinguished by the fact that their somas are inside the EM imaging volume; as such, they are generally more completely reconstructed compared with other neurites and have significantly more synapses.

(legend continued on next page)

(all ~3.2 million of them, besides the 5,623 annotated synapses between cells with somas; that is, less than 0.2% of all synapses) are formed between neurites that cannot be assigned a valence from morphology, precisely because their somas are not inside the imaged volume. The result is that >99.8% of synapses in the dataset are not being used to describe circuitry characteristics.

In the previous section, we showed that for individual synapses, SynapseCLR representations can be used to reliably predict pre- and postsynaptic cell types (0.975 and 0.940 AUC, respectively). Here, we exploit the fact that a typical synapse-forming neurite in this dataset that is not connected to a soma contains tens of synapses (for neurites connected to a soma, this figure reaches thousands of synapses; [Figure 4C](#)). Therefore, even with imperfect cell typing for individual synapses, our ability to confidently assign a cell type to each neurite will be dramatically enhanced (with errors decreasing exponentially with the number of synapses per neurite) by taking a consensus of all participating synapses in a maximum likelihood estimation fashion. The consensus cell type of each neurite can then be propagated back to the participating synapses ([Figure 4A](#) and section “experimental procedures”).

We evaluated the accuracy of our consensus cell-typing algorithm by cross-validation: we randomly split the available annotated synapses into training and testing sets, trained single-synapse cell type classifiers on the training set, performed consensus cell-type calling on all synapses (annotated and unannotated), and evaluated the accuracy of consensus cell-type calls over the held-out test set. We repeated the random train/test splitting five times to obtain credible intervals, and repeated the entire procedure for different train/test splitting ratios in order to study the impact of training set size on the generalization accuracy of consensus cell type calls. We refer to the percentage ratio of the training set size to all available synapses as annotation completeness. Finally, we stratified the cross-validation accuracies of consensus calls by actual pre- and postsynaptic cell types and by several bins of the number of synapses per neurite (SpN) in order to study how many synapses might be required for significant consensus-based improvement in cell-typing accuracy compared with cell typing on the basis of individual synapses ([Figure 4B](#)).

We found that excitatory neurites could be typed with virtually 100% accuracy using consensus calling, even at 1% annotation completeness (corresponding to a training set size of 950 annotated synapses, the lowest level we tested), and with $\text{SpN} \leq 20$ ([Figure 4B](#), top panels). We found consensus cell type accuracy to be similar for inhibitory presynaptic neurites ([Figure 4B](#), bottom left panel), but the accuracies were somewhat lower for inhibitory postsynaptic neurites. We attributed these lower accuracies to the lower prevalence of inhibitory neurons with somas in this EM dataset (14, compared with 364 excitatory neurons with somas). This lower prevalence translates to a relative scarcity of annotated synapses formed onto inhib-

itory neurons (1,108 out of 5,623, compared with 4,515 synapses onto excitatory neurons; see [Figure 4D](#), left) and thus, a smaller slice of the training dataset that represents this post-synaptic cell type. Nevertheless, virtually 100% consensus cell-typing accuracy was achieved for $\geq 4\%$ annotation completeness.

By studying the distribution of SpN in the full EM dataset ([Figure 4C](#)), we estimated a lower bound for the mean consensus cell-typing accuracy over the full ~3.2 million detected synapses. To this end, we took our least accurate category of single-synapse prediction (postsynaptic inhibitory, $p = 0.846$ at 5% annotation completeness; see [Figure 4B](#), bottom right), and assumed all single-synapse predictions to exhibit a similar accuracy. The probability of successful consensus cell typing per neurite for a given SpN can be calculated analytically from the binomial distribution. Explicitly, we take SpN as the number of trials, successful classification probability $p = 0.846$, and calculate the total probability of having at least $(\text{SpN}/2) + 1$ correct trials. We calculate the mean value of the resulting probability over the empirical distribution of SpN in our dataset ([Figure 4C](#)), and find a mean lower-bound consensus call accuracy of 0.998.

Finally, as a consistency check, we trained the regression models on all of the available annotated synapses and ran the consensus calling algorithm on those same synapses. The resulting consensus cell types were reassuringly in perfect agreement with the actual annotations (see [Figure 4D](#), middle). Finally, we ran the consensus calling algorithm over the entire set of synapses in order to learn what kind of neurites they were connecting and by extension, to learn about the distribution of synapse types in the dataset. We found that E-to-E synapses dominated the dataset, forming 82% of our sampled synapses (77,622 out of 94,874). Overall, excitatory synapses (E-to-E plus E-to-I) formed 88% (83,133 out of 94,874) of synapses in the volume, consistent with previous estimates of the balance of excitatory versus inhibitory synapses in cerebral cortex⁴; [Figure 4D](#), right).

SynapseCLR representations reveal the intrinsic dimensionality of synaptic structural variation and capture neuronal-subtype-specific features

The accuracy with which our representation regression models can predict complex and fairly abstracted synapse annotations, such as neuronal type, hints at the biologically informed nature of these representations, and encourages us to study the manifold structure of these representations per se. In particular, we are interested to learn about the intrinsic dimensionality and independent axes of synaptic structural variation, to the degree and granularity that is allowed by the EM data, and through the lens of SynapseCLR representations. These data-driven axes of variation will provide a clearer picture of the isolated synapse, without dependence on human-derived annotations (many of which strongly co-vary and are not independent, while others might involve information that is not contained in the isolated synapse).

(D) The distribution of pre- and postsynaptic cell types for (left) manually annotated synapses, (middle) consensus cell type calls restricted to manually annotated synapses, and (right) consensus cell type calls over all 94,874 randomly chosen synapses from our dataset (L2/3 mouse primary visual cortex). Note the bias in the manual annotations (they are significantly enriched in inhibitory synapses), the perfect agreement between manual annotations and our consensus cell type calls, and the unbiased cell type calls when synapses are chosen randomly from L2/3 primary visual cortex, in good agreement with previous reports of the excitatory-to-inhibitory synapse ratio.

As noted earlier, the 2D UMAP embedding of SynapseCLR representations contains two connected components, one roughly corresponding to E-to-E synapses (see [Figures 2G](#) and [2I](#), the upper-left component) and the other corresponding to I-to-E/E-to-I/I-to-I synapses (see [Figures 2G](#) and [2I](#), the lower-right component). We found this global compartmentalization to be robust using a variety of clustering techniques, including density-based spatial clustering of applications with noise (DBSCAN)⁷⁴ density-based clustering and Leiden⁶⁹ community detection. Therefore, we set on to study the two connected components separately, reasoning that they harbor structurally distinct synapses.

The exercise of mapping the axes of smooth variation and the low-dimensional structure that may underscore high-dimensional data (here, our 512-dimensional SynapseCLR representations) is referred to as manifold learning and is an area in machine learning that has witnessed rapid progress in recent years. A widely used method to this end is the principal-component analysis (PCA) and operates under the assumption that the data manifold is Euclidean (flat). Here, there is no *a priori* indication that our representations admit a Euclidean structure. In fact, we find that many principal components (PCs) are needed to capture the variance of the representations, suggesting that the data manifold is not Euclidean (see [Figure S10B](#)). By contrast, the previously noted smooth variation of structural annotations over 2D UMAP embedding provides a strong support for the existence of a low-dimensional manifold, albeit non-Euclidean (curved). We remark that, even though UMAP is most frequently used as a visual aid for high-dimensional data, it is in fact a general-purpose and rigorous manifold learning framework, is applicable to both linear and non-linear spaces and operates under minimal assumptions⁶⁸ (global connectedness and uniform distribution of points on the manifold). As such, we decided to use UMAP as the manifold learning algorithm and devised a three-step strategy to learn and interpret the manifold structure as follows.

Manifold learning step

First, we isolated the synapses contained in each connected component of the representation space and generated UMAP embeddings of their representations into n -dimensional target spaces. We varied n from 2 to 10 in order to identify the smallest n needed to embed the data faithfully (see below).

Manifold parameterization standardization step

Given that UMAP embeddings are arbitrary up to certain global transformations (i.e., translation, reflection, and rotation) and that these factors are determined by random initialization of the graph layout optimization problem solved by UMAP,⁶⁸ we performed PCA on the UMAP embeddings in order to standardize the parameterization. We kept all n principal components and the application of PCA was not for the purpose of further dimensionality reduction; rather, we leveraged PCA as a means to rearrange the UMAP embedding space (i.e., center and rotate) along directions that explain variance of the learned manifold in descending order. We refer to this procedure as UMAP-PCA and refer to the obtained manifold parameters as UMAP-PCs. As the target manifold dimension n is increased, UMAP embeddings become increasingly faithful to the structure of the data manifold. At the same time, we observe that additional UMAP-PCs explain rapidly diminishing amounts of vari-

ance. We identify the smallest n such that UMAP-PC($n + 1$) explains less than 5% of the manifold parameterization variance as the approximate intrinsic dimensionality of the manifold.

Manifold interpretation step

Finally, we study the correlations between the loadings of UMAP-PCs and the annotations in order to interpret these axes of variation in terms of known anatomical measurements.

The result of the 2D UMAP-PCA analysis is shown in [Figure 5](#) (top row, excitatory component; bottom row, inhibitory component). To guide the eye, each component has been color coded such that the UMAP-PC 1 axis has a white-to-red gradient and the UMAP-PC 2 axis has a white-to-blue gradient ([Figures 5A](#) and [5E](#)). Examples from a uniformly sampled 2D UMAP-PC grid ([Figures 5B](#) and [5F](#)) suggest smooth changes of multiple anatomical features along each direction (see also [Figures S6](#) and [S7](#)). The Spearman correlation between the UMAP-PC loadings and the annotations are shown in [Figures 5C](#) and [5G](#), together with the ratio of explained variance (EV) by each axis. As [Figure 5C](#) shows, UMAP-PC 1 explained more than 96% of the variance of E-to-E synapses, suggesting that the structural variation captured by SynapseCLR for these synapses, by and large, can be explained by a single degree of freedom. Moving along this axis is positively correlated with increasing pre- and postsynaptic partner volume, synaptic cleft size (which is limited, but not determined, by pre- and postsynaptic partner volume), and presynaptic mitochondrion presence and volume. The remaining 4% of variation is largely explained by UMAP-PC 2, and extending the analysis to higher dimensions does not change this finding (see [Figure S8A](#)). This second PC correlates with increased presynaptic partner volume; presynaptic mitochondrion presence and volume; and, notably, the probability that the presynaptic partner is inhibitory. Indeed, [Figure 3B](#) shows that UMAP-PC 2 is directed toward the region of this component where I-to-E synapses are concentrated.

In the lower component, which contains mostly synapses involving inhibitory neurons, we find that again a large fraction of the variation (80.4%) in representations is explained by UMAP-PC 1. Compared with the excitatory component, this reduced amount of variance explained by a single axis suggests that this population of synapses is structurally more diverse; this observation is consistent with previous documentation of greater structural diversity among inhibitory neurons and their synapses relative to excitatory ones in mammalian cortex.^{2,75,76} As with the excitatory component, repeating this analysis in higher dimensions does not significantly reduce the relevance of the top two UMAP-PCs, suggesting that the inhibitory synapses admit two main axes of variability (see [Figure S8B](#)). In UMAP-PC 1, we observed two relationships of note: first, the probability of the presynaptic partner being inhibitory increases along UMAP-PC 1, whereas the probability of the postsynaptic partner being inhibitory decreases along this axis. Second, the distance to the postsynaptic soma decreases along UMAP-PC 1. The switching of synapse type from E-to-I to I-to-E along this axis, combined with a systematic movement of synapses formed closer to the postsynaptic soma, suggested that maybe this axis captured partitioning of synapses formed between different cell subtypes. UMAP-PC 2 correlated with several annotations, but the mixture did not afford us a clear interpretation on the basis of the anatomical properties that might be varying along this

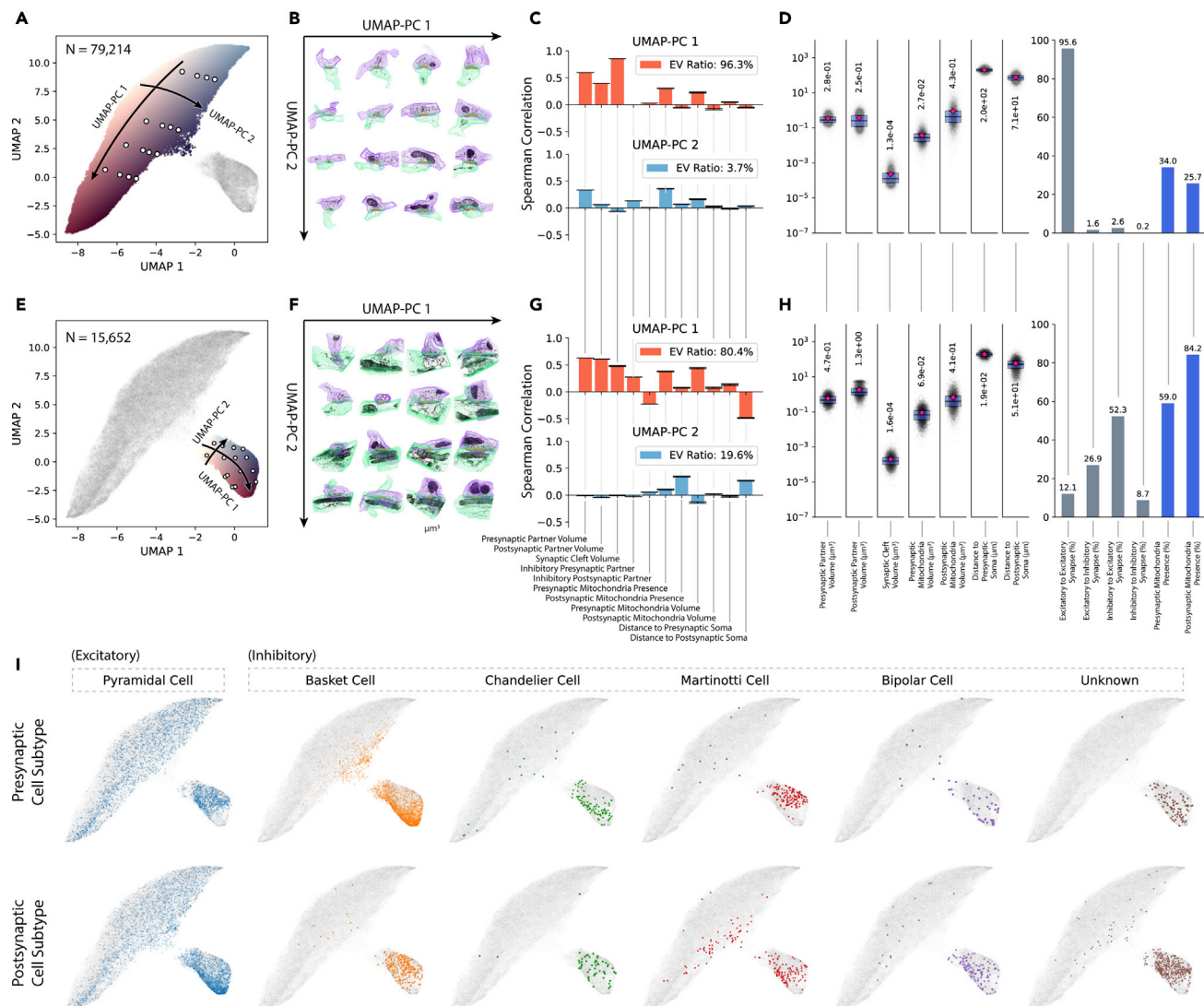


Figure 5. Inhibitory and excitatory synapses form disjoint, low-dimensional, smooth manifolds in the SynapseCLR representation space

(A) UMAP embedding highlighting the excitatory (upper left) component, colored by the coordinates of our 2D UMAP-PCA analysis (see section “[experimental procedures](#)”). The directions of UMAP-PC 1 and UMAP-PC 2 are marked by colored gradients of white to red and white to blue, respectively. The white circles correspond to the synapses shown in (B). The black arrows are guides for the eye.

(B) Synapses along the 2D UMAP-PCA grid of the excitatory component, showing visible feature changes, including increasing synaptic cleft size along UMAP-PC 1 and presynaptic mitochondria presence along UMAP-PC 2.

(C) Spearman correlation of various annotations versus UMAP-PCs in the excitatory component. The legend shows the ratio of EV of the UMAP manifold parameterization by each UMAP-PC.

(D) Summary statistics of continuous annotations (left) and discrete annotations (right) for all synapses belonging to the excitatory component. Boxes correspond to the inter-quartile range; red markers and numerical values denote the mean.

(E) Similar to (A) but for the inhibitory (lower right) component.

(F) Similar to (B) but for the inhibitory component; note increasing synaptic cleft size and postsynaptic partner volume along UMAP-PC 1, and increasing postsynaptic mitochondria presence along UMAP-PC 2.

(G) Similar to (C) but for the inhibitory component.

(H) Similar to (D) but for the inhibitory component.

(I) Pre- and postsynaptic neuronal subtypes shown over a subset of annotated synapses, showing that synapses formed by different neuronal subtypes appear in different parts of the representation space.

axis. However, the variation along UMAP-PC 1 led us to wonder whether UMAP-PC 2 also might be capturing cell-subtype-specific variation in synapse structure. To test this, we highlighted cell subtypes for the subset of annotated synapses that had

them (Figure 5I). All excitatory neurons were labeled “pyramidal” (because no structural subtypes could be easily identified); inhibitory neurons with substantial axons (20) were classified previously as basket, chandelier, Martinotti, or bipolar neurons, or

were otherwise labeled as having an unknown subtype.² We found that the lower component stratified discernibly according to these subtypes. Along UMAP-PC 1, synapses went from being formed by pyramidal onto a mix of inhibitory subtypes (largely basket neurons) to being basket-to-pyramidal synapses (Figure 5I, top row). Notably, basket neurons form synapses directly onto pyramidal neuron somas, which is consistent with our finding of decreasing distances to the postsynaptic soma in this region (Figure 3B). Along UMAP-PC 2, inhibitory neurons that formed synapses onto other cell types were remarkably separated according to cell subtype, stratifying into synapses formed by basket, then chandelier, then Martinotti presynaptic neurons. Synapses with bipolar presynaptic neurons also appeared to occupy multiple but specific regions of this component; this observation may reflect the fact that there are different structural subtypes of bipolar neurons in this part of the visual cortex.⁷⁶ By contrast, the representations of synapses formed onto inhibitory neurons of different subtypes overlap substantially (Figure 5I, bottom row), indicating that inhibitory dendrites may not differ as much for different inhibitory subtypes at the spatial scale of the synapse.

The correlations we highlighted in this section via manifold learning could have been established more directly by calculating pairwise correlations in the available annotations. Such an analysis is shown in Figure S3 and is in qualitative agreement with the UMAP-PCA analysis. We would, however, like to emphasize that, had the annotation set been smaller, a direct analysis would have failed to provide statistically significant correlations. By contrast, the manifold-aided correlation analysis requires a much smaller annotation set: having the manifold structure at our disposal, just a few annotated synapses will be sufficient to reveal the correlations, owing to the smooth variation of annotations over the manifold. Thus, this semi-supervised approach to analysis, where the available annotations are used to help interpret the low-dimensional manifold structure of self-supervised representations, may help reduce the manual annotation burden in future studies.

Finally, we asserted the robustness and reproducibility of the results presented in this section by repeating the analysis using several different UMAP random initializations and by varying UMAP key parameters (min_dist, spread) over an extended range (0.1–0.3 and 1.0–5.0, respectively). In all instances, we obtained the same UMAP-PCs and EV ratios within 5% relative distance of the results shown here (corresponding to n_neighbors = 1,000, min_dist = 0.1, spread = 1.0; cuML UMAP implementation⁷⁷).

DISCUSSION

Summary

In this work, we have presented a self-supervised contrastive representation learning method called SynapseCLR, an adaptation of the SimCLR framework to 3D EM data, and used the method to learn feature representations of synapses in a 3D EM dataset from mouse visual cortex. We established that our representations separated synapses according to both their overall physical appearance and structural annotations of known functional importance. We demonstrated the utility of our methodology for several valuable downstream tasks for the growing

field of 3D EM connectomics. These include one-shot identification of defective synapse segmentations, dataset-wide similarity-based querying, and accurate imputation of annotations for unlabeled synapses, using only manual annotation of 0.2% of synapses in the dataset. In particular, we show that neuronal cell types (excitatory versus inhibitory) can be assigned to individual synapses and highly truncated neurites with accuracy exceeding 99.8%, making this population accessible to connectomics analysis. Finally, we presented a data-driven and unsupervised classification of synaptic structural variation, revealing its intrinsic dimensionality and showing that synapse structure as captured in SynapseCLR representations also contains information related to inhibitory neuronal subtypes.

SynapseCLR representations solve multiple outstanding issues with large-scale 3D EM connectomics data analysis

Our proposed approach to studying synapses is useful for several reasons. First, it is a more scalable, less biased way of making sense of neural circuitry components compared with manual annotation and the conventional supervised machine learning approaches. To learn representations with SynapseCLR, only cell-level segmentations for synapses of interest are needed, as is an understanding of technical sources of variation among images (e.g., variation in pixel intensity, rotation, sectioning artifacts). This approach reduces the dimensionality of raw synapse images (which can be millions of voxels worth of information) to a few-hundred-dimensional feature space that is, by design, optimally discriminating.

Working with low-dimensional feature representations has two key practical advantages. One is that having a relatively small number of annotated synapses (several thousands) is sufficient to train accurate predictive models without overfitting. We recall that the dimensionality of backbone features (512) is an order of magnitude smaller than the number of annotated synapses (5,623), allowing training regression models without overfitting. By contrast, the dimensionality of raw EM image chunks ($96 \times 96 \times 96 \approx 10^6$) is nearly two orders of magnitude larger than the number of annotated synapses, posing a fundamental challenge for supervised learning. Liberation from intensive manual data labeling has been extensively studied in the context of classifying natural images and is the main motivation behind the development of various self-supervised pretraining regimens.⁵⁵ The other advantage of working with low-dimensional feature representations is the possibility of using unsupervised machine learning methods to explore and interpret these representations per se. For example, unsupervised manifold learning using UMAP showed us that representations may split according to neuronal subtype: this finding reinforces previous descriptions of the subtype-specific structural differences that exist among synapses (in terms of, e.g., synaptic target cell types, target synapse locations and geometries, and neurotransmitter-related machinery^{35,75,76,78}), and, importantly, it suggests that SynapseCLR representations encode information about such structural elements. Future directions include highlighting additional organelles of interest in further explaining synaptic structural variation between neuronal subtypes and linking this to neuronal microcircuits. More broadly, application of unsupervised methods and de-emphasis of human-designed features as

afforded by SynapseCLR will be essential for circuit property discovery as EM image volumes continue to grow in size.^{3,20,41,79}

Second, our representations capture not only biologically important variations in synaptic structure but also non-specific, technically important variations in structure that we did not factor into our augmentations (i.e., segmentation artifacts). Defective synapses can be identified and removed in one shot using only a handful of examples, as shown in [Figures 2A–2F](#). This characteristic of SynapseCLR representations is reminiscent of few-shot learning approaches in computer vision,^{60,80} and is of great value for quality controlling and analyzing large connectomics datasets, where otherwise full, manual inspection of synapses would be required to ensure that the synapses being studied are not contaminated.

Third, SynapseCLR representations capture features that in particular define pre- and postsynaptic cell type (E versus I; [Figure 2I](#)) with high accuracy. By ensembling single-synapse predictions over individual neurites, we were able to assign highly accurate cell types for the many highly truncated neurites in this EM dataset that formed most of our unannotated synapses ([Figure 4](#)). This capability provides huge value to the field of connectomics, because usually cell types can only be assigned to mostly complete neurons (or glia) in a dataset which comprise a small minority of neurites (here, 400 of 8 million, or $5 \times 10^{-3}\%$, of objects) and the majority of objects are left out of the analysis. By running imputations and making consensus calls on neurites that form two or more synapses, we estimate that we are able to put ~378,000 of them back into our analysis with ~100%-accurate cell types (roughly 30% of the 1.27 million neurites that form putative synapses in the volume, and 4.7% of the 8 million cellular segments in the volume), significantly enhancing future studies of cell-type-specific connectivity patterns.

Appropriate augmentation design is critical for finding meaningful self-supervised representations

The learning paradigm used in SynapseCLR is designed to retain the most robust discriminating visual features that remain invariant under a specified set of augmentations while distilling out non-invariant visual data. Thus, in order for SynapseCLR to generate biologically meaningful synapse representations, augmentation choice is critical: augmentations should be designed to capture as much technically sourced image variation as possible and to avoid capturing biological variations. We motivate our augmentation choices in great detail in the “[experimental procedures](#)” section. As a concrete example of the way insufficient augmentation can degrade representations, we trained SynapseCLR a second time with a modified type of rotation. With our original approach, where full SO(3) rotations (i.e., arbitrary rotations in 3D space) were allowed (giving the results in [Figures 2–5](#)), we see that the representation space is indeed invariant to synapse orientation angle ([Figures S9A and S9C](#)) while successfully capturing variation in other structural properties (see section “[results](#)”). (One exception is the subset of E-to-E synapses that occupy the upper-right region of the upper manifold component; this may be indicative that synapse orientation is highly conserved for synapses formed by some excitatory cell subtypes; see below.) In contrast, when we restricted augmentations to discrete octahedral rotations only, the resulting representations were dominated by synapse orientation ([Fig-](#)

[ure S9B](#)). As a control demonstrating that our augmentation procedure is functioning properly, the number of blanked sections was varied in augmentations for both runs, and in both the resulting representations had no dependence on the number of missing sections ([Figures S9D–S9F](#)).

SynapseCLR representations suggest new directions for future studies of neuronal circuit connectivity

The representations generated by SynapseCLR show that structural variations among synapses correspond to notable annotation properties that should be further investigated. First, by studying the axis of greatest variation for E-to-E synapses (the long axis of the upper component; [Figure 5C](#), UMAP-PC 1) we see that the primary way in which synapses differ along this axis is by cleft size, and it appears that synapses with larger clefts have more presynaptic mitochondrial presence and less postsynaptic mitochondrial presence than smaller synapses. These relationships could be a marker for E-to-E synapses that are being potentiated: when a pre- and postsynaptic cell repeatedly fire in a fixed temporal pattern, with presynaptic firing followed tens to hundreds of milliseconds later by postsynaptic firing (this is also called synchronous firing), synapses are thought to undergo changes that mediate more efficient signal transmission.^{81,82} One such change for excitatory neurons over longer times (roughly hours or longer) is addition of α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors (AMPA receptors)⁸², which mediate signal transmission for the excitatory neurotransmitter glutamate; this would correspond to a larger synaptic cleft in an EM image, since cleft size is typically measured as the volume of the postsynaptic density (the net of architectural proteins and receptors positioned on the postsynaptic side of the synapse³⁰). Mitochondria may also have a role in promoting synaptic potentiation: in addition to providing cells with energy (via oxidative phosphorylation that produces ATP), mitochondria buffer Ca^{2+} out of surrounding cytosol.⁵⁰ Ca^{2+} is essential for the release of vesicles containing neurotransmitters from the presynaptic cell, and one consequence of reduced levels of presynaptic Ca^{2+} is reduced spontaneous vesicle release. Mitochondria-mediated removal of presynaptic Ca^{2+} suppresses such asynchronous activity and promotes synchronous firing leading to potentiation.⁸³ In the postsynaptic region, mitochondria are the primary energy source responsible for local translation of mRNA into proteins.⁵⁰ It has been proposed that enhanced postsynaptic mitochondrial presence may be a signature of synaptic downscaling, which is in turn a consequence of synaptic potentiation: broad postsynaptic downscaling allows potentiated synapses to take larger weights in signal transmission without triggering runaway activity (epileptic events).^{84,85} Future work could focus on determining whether these distinctions do in fact indicate a subpopulation of potentiated E-to-E synapses.

Second, as highlighted by our UMAP-PCA results and as mentioned above, representations appear to separate synapses not only by pre- and postsynaptic cell type but also by cell subtype (as we see from our synapses involving inhibitory neurons; [Figure 5I](#)). This finding is important because it suggests that neurites can be assigned cell subtypes based on the structural representations of the synapses they form. In the future, annotations of these synapses could be used to impute cell subtypes for neurites in EM datasets and thus add a layer of specificity to

connectivity analyses. This type of information could further build out the picture of cortical connectivity, by highlighting the frequency of various cell connection types and describing the synapse properties of those connections.

The finding that inhibitory neurons can be subtyped based on their synaptic representations begs the question of whether the same can be said about excitatory neurons. As shown in [Figure S9A](#), the E-to-E component does possess a region where synapse orientation is important (the upper-right region). This region could either correspond to synapses involving a specific compartment of an excitatory neuron (pyramidal cells have highly stereotyped dendritic morphologies, including apical dendrites that point radially and basal dendrites directed 45° away from the apical axis), or it could correspond to a specific subtype of excitatory neuron. Further work to identify annotations that might identify excitatory cell subtypes (such as compartment labels or cortical layer labels; e.g., L2/3 versus L5/6) would be useful for learning more about connectivity properties of excitatory subtypes (which are notoriously difficult to subtype based on 3D EM morphology alone).

Additionally, the finding that E-to-E synapse representations may separate based on potentiation state begs the question of whether this is true for inhibitory neurons. Indeed, this and many other facets of the synapses in this dataset can be investigated by identifying additional structural correlates of function. For example, synaptic vesicles (both docked and free⁸⁶) and endoplasmic reticulum (ER)³⁶ are two organelles already known to be essential for synaptic function that could be annotated to help interpret SynapseCLR representations. One of the salient characteristics of contrastive learning is the implicit inversion of the data-generating process (here, meaning aspects of cellular function) by learning disentangled representations at the projection head.^{87,88} Here, a disentangled feature means one that controls a specific aspect of the synapse structure while remaining inside the data manifold and leaving other aspects intact (i.e., being an independent knob). We have empirically demonstrated this emergent property for SynapseCLR representations in [Figure S10A](#), where projection head features are found to be largely independent of one another. One may speculate that disentangled visual features of synapses may correspond to biological function. Indeed, we find that projection head PCs 1 and 8 control cleft size and mitochondrial presence (although not independently; see [Figure S10C](#)), both features being of known functional significance. Projection head PC 3, on the other hand, while being largely uncorrelated with our available manual annotations, seems to be responding to synaptic vesicles upon visual inspection ([Figure S10C](#)). More generally, we speculate that relating SynapseCLR representations to structural correlates of function, paired with visual assessment of synapses showing high and low activation of these features, may reveal novel anatomical properties of synapses tied to their function.

SynapseCLR representations themselves can be enhanced by several future endeavors. First, training on larger input image chunks could allow for more relevant information over larger spatial scales to be encoded. Such information could include how synapses may vary with distance from cell somas (in this work, we trained on images spanning a few micrometers, whereas dendrites and axons extend for hundreds of micrometers and up to 400 μm in this dataset²). It could also allow for more informative

encoding of mitochondrial features by capturing more of the mitochondria near synapses: in our image chunks, mitochondria were often truncated because they were either much larger than the volume (true for most postsynaptic mitochondria) or because they were slightly farther from the synaptic cleft than the image chunk boundaries (a source of noise for both pre- and postsynaptic mitochondria). Similarly, training with a larger network to produce SynapseCLR representations could lead to more detailed representations, as has been shown by past work.^{53,56}

SynapseCLR can be readily used for learning representations from other objects in an EM dataset besides synapses. Correlating synapse representations with the representation of the surrounding neuropil is an attractive future direction for revealing potential modulation of synapse structure by nearby processes (e.g., microglia and other glial cell types are known to modulate synaptic function^{38,66}).

Looking farther into the future, generation and interpretation of unsupervised representations in EM could be further enhanced by adding multimodal information, such as focused, higher-resolution EM; molecular image labels through immunohistochemistry or by other means; or activity-based images using techniques such as Ca²⁺ imaging. Specific targets for either input labels or annotations could include electrical synapses⁸⁹ or more subtle synaptic organelles such as multivesicular-endoplasmic reticulum bodies.⁹⁰ Given that transcriptional states are strongly correlated with morphological features in cells and tissues,^{91–94} we believe that informative morphological representations such as those produced with SynapseCLR may help form a more complete picture about the connection between structural connectivity and the underlying molecular mechanisms.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Alyssa Wilson (alyssa.wilson@mssm.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- This paper analyzes existing, publicly available data hosted at <https://www.microns-explorer.org/phase1> (DOI: <https://doi.org/10.5281/zenodo.7510511>). The derivatives of the data used in this paper (unmodified and processed chunks of 3D EM images and segmentations) and original data in this paper (pretrained SynapseCLR models, SynapseCLR representations, analysis outputs) have been deposited to Terra: <https://app.terra.bio/#workspaces/broad-firecloud-dsde/SynapseCLR>. These data are publicly available as of the date of publication.
- All original code has been deposited to Zenodo: <https://doi.org/10.5281/zenodo.7545417>. This code is publicly available as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EM source dataset

Our 3D EM image volume was a reconstruction from layer 2/3 (L2/3) of primary visual cortex in a young-adult (postnatal day 36), male mouse (from the IARPA MICrONS project; see <https://www.microns-explorer.org/phase1>). This volume was built from ~2,250 EM images, each 40 nm thick and with resolution 3.58 × 3.58 nm², and measured 240 × 140 × 90 μm³ in total. We made use of three segmentation layers that were also released with this dataset: one layer

segmenting the EM images into its constituent cells; one identifying the active regions of chemical synapses (called synaptic clefts) in the volume; and one showing all the mitochondria present in the volume. We also used skeletons for 397 excitatory and inhibitory neurons with their somas in the volume. These skeletons had been proofread and smoothed to allow for calculations involving distances along the dendrites or axons of these neurons.

EM and segmentation mask image chunk generation

To generate image tensors for training, we pulled the list of synaptic clefts published with the MICrONS L2/3 mouse primary visual cortex dataset.⁹⁵ This list contains 3.2 million rows of putative synapses and a subset of 6,761 synapses that had been previously proofread by multiple experts. These synapses were formed between excitatory and inhibitory neurons that had their somas contained inside the dataset and were used for multiple analyses.^{2,39,75}

We then sampled additional synapses to generate a targeted training dataset of 100,000 synapses by randomly sampling from the rest of the synapse list (which contained synapses scattered effectively uniformly throughout the EM dataset). For every synapse we chose, we downloaded a $256 \times 256 \times 256$ -voxel EM image chunk at MIP 1 ($7.16 \times 7.16 \times 40 \text{ nm}^3/\text{voxel}$) from the public data repository, and only added it to our input image if $<10\%$ of the image chunk was missing (i.e., masked out to zeros in order to prevent section collection or imaging artifacts from affecting image volume post-processing). If a chunk had more than 10% of its volume masked out, it was discarded from the input set and a different synapse was selected and checked for suitability.

If the synapse met our criteria for EM masking, we also downloaded chunks of the same size and centered at the same location from the segmentation, synaptic cleft, and mitochondrial layers, to be used for input tensor generation and, for synapses connecting neurons with somas in the volume, annotation population. Input image data were downloaded for all synapses using CloudVolume⁹⁶.

For the segmentation layer, we restricted our download to only the pre- and postsynaptic neurons forming the current synapse, using the synapse table provided with this dataset. If this process failed for any of the synapses for which annotations were possible, we discarded it from our training set. Because we also used mitochondrial segmentations for our annotation-based analysis, we also discarded synapses where the mitochondrial segmentation was not available. This process resulted in a training dataset consisting of 5,664 annotated and 91,994 unannotated synapses.

Keeping in mind that we ultimately wanted to work with isotropic images, we kept only the 52 central z sections that covered roughly the same physical linear dimension as x and y (i.e., 52 sections at $40 \text{ nm}/\text{px} = 2,080 \text{ nm}$, 256 pixels at $7.16 \text{ nm}/\text{px} = 1,832 \text{ nm}$). During data augmentations, we then upsampled our z pixels to roughly match our x and y resolutions (see below).

Volumetric data augmentation

We constructed a set of intensity-, masking-, and affine-transform-based augmentations to use for training our network on synapse image data (Figure 1C). Each transformation is defined, first, by a probability parameter, which fixes the probability at which it will be included in an instance of an augmentation, and, second, by hyperparameters that define the exact orientation, shape, or magnitude of the transform.

We implement augmentation transformations in PyTorch⁹⁷ to leverage GPU acceleration. Transforming volumetric data is considerably more time consuming than 2D images and the conventional CPU-based approach to data augmentation results in considerable slowdown and wasted GPU time. The overall data flow for augmenting a batch of n volumetric image chunks is as follows.

1. The volumetric intensity data are transformed to $[0, 1]$ value range and are represented as a batch of 3D tensors of size $256 \times 256 \times 52$, and the binary segmentation masks (presynaptic process, synaptic cleft, postsynaptic process) are represented as a batch of three-channel 3D binary tensors of size $256 \times 256 \times 52$. We note that, at this stage, the representation is anisotropic: each voxel corresponds to a physical volume of $7.16 \times 7.16 \times 40 \text{ nm}^3$.
2. A first series of intensity-based and mask-based transformations are randomly sampled and applied to the intensity and the mask tensors, respectively. These transformations are described below.

3. An affine transformation composed of a random 3D rotation, random center displacement, central cropping, and a fixed anisotropic scaling is applied to both intensity and mask tensors. Central cropping and anisotropic scaling transformation plays three roles: (1) reverting the unequal sectional and in-plane tissue imaging resolution and ensuring that the final voxels are isotropic at $7.16 \text{ nm}/\text{px}$ a side; (2) shrinking the field of view from $\sim 2 \mu\text{m}^3$ (256 px) to $\sim 1.5 \mu\text{m}^3$ (192 px) in order to reduce the external context; (3) downsampling the resolution to the final desired output resolution of $96 \times 96 \times 96$ voxels, with each voxel corresponding to a physical volume of $14.3 \times 14.3 \times 14.3 \text{ nm}^3$. We implement this series of transformations efficiently by leveraging PyTorch's Spatial Transformer Network (SPN) capabilities as a single operation.⁹⁸
4. A second series of intensity-based and mask-based transformations are randomly sampled and applied to the isotropic intensity and the mask tensors, respectively. These transformations are described below.
5. Finally, the three-channel mask tensor is merged to a single binary mask via a voxel- and bit-wise OR operation. Peripheral voxels that may be disconnected from the synaptic cleft are identified and removed using the cc3d package⁹⁹ by traversing the 26 nearest neighbors of each voxel. The intensity channel is multiplied with the resulting active zone binary mask, z scored using dataset-wide mean and SD, and returned.

We describe each transformation and our choice of hyperparameters below.

Intensity-based transformations

Sectional cutout

A common technical artifact of 3D EM datasets generated via serial sectioning is the random missingness of parts of many scanned sections due to tissue sectioning or scanning issues (e.g., thin tissue deformation, warping, folding). A typically employed QC measure is to identify and blank out affected regions along the fold line. Since the blanked-out regions can be tens of micrometers in width, these partially missing regions appear as completely missing sections when sampling micrometer-sized 3D volumes (e.g., a synapse) from the reconstructed dataset. Such a prominent technical artifact may be easily exploited by the representation learner to trivially solve the contrastive objective without learning a deeper structural feature representation. We estimated a typical synapse to have 1–2 missing sections in our dataset. To disincentivize the model from incorporating such artifacts into the representation space, we randomly select $n_{\text{cutout}} \sim \text{Poisson}(10)$ sections and blank them out. This transformation is applied before the affine transformation and with probability 0.5.

Sectional intensity distortion

The EM dataset is normalized appropriately to minimize sectional intensity variations. The normalization strategy implemented by the original authors, however, is not locally adaptive and, thus, we estimated approximately 10% sectional intensity variation. To desensitize the representations to such technical variations, we apply following transform stochastically to every local z section:

$$I_{\text{out}} = \alpha + \beta I_{\text{in}}^{\gamma} \quad (\text{Equation 1})$$

where I_{in} and I_{out} denote the input and output z sections, respectively. The transformation is applied pixel-wise, and, for each z section, α , β , and γ are independent and identically distributed (i.i.d.) random variables sampled as follows:

$$\alpha \sim \text{Uniform}(-0.05, 0.05) \quad (\text{Equation 2})$$

$$\beta \sim \text{Uniform}(0.9, 1.1)$$

$$\gamma \sim \text{Uniform}(0.9, 1.1)$$

The output intensities I_{out} are clipped to range $[0, 1]$ to prevent under- and overflows. This transformation is applied before the affine transformation and with probability 1 (always).

Global intensity distortion

A similar reasoning applies to technical intensity variations at the volumetric level; i.e., presence of intensity variations at the scale of tens of micrometers

due to varying tissue quality, scanning batch, etc. As a remedy, we apply the same transformation as above to the entire 3D volume:

$$V_{out} = \alpha + \beta V_{in}^{\gamma} \quad (\text{Equation 3})$$

where V_{in} and V_{out} denote the input and output volumes, respectively. The transformation is applied pixel-wise, and α , β , and γ are sampled as before. In contrast to sectional distortion, α , β , and γ are not independent across z sections and are sampled once for the entire volume. The output intensities V_{out} are clipped to range [0, 1] to prevent under- and overflows. This transformation is applied before the affine transformation and with probability 1 (always).

Pixel noise

Independent pixel noise is a ubiquitous source of noise in any imaging. We apply both normal additive and log-normal multiplicative independent noise to every voxel:

$$V_{out} = A + M V_{in} \quad (\text{Equation 4})$$

where A and M (positive) are 3D additive and multiplicative i.i.d. noise tensors with the same dimensionality as V_{in} . We estimated the pixel noise level to be approximately 1%–5% by comparing the intensity of adjacent pixels. Moreover, we observed variation in the level of noise at the scale of tens of micrometers (likely due to varying tissue condition). We model this using a simple two-hierarchy noise sampling procedure:

$$\sigma_A \sim \text{Uniform}(0, 0.05) \quad (\text{Equation 5})$$

$$\sigma_M \sim \text{Uniform}(0, 0.05)$$

$$A \sim \text{Normal}(0, \sigma_A)$$

$$M \sim \text{LogNormal}(0, \sigma_M)$$

The output intensities V_{out} are clipped to range [0, 1] to prevent under- and overflows. This transformation is applied before the affine transformation and with probability 1 (always).

Gaussian blur

Another source of imaging variation is blurring, which, in EM datasets, stems from variation in tissue quality and dye diffusion. While blurring is not considered a significant issue for the scanning EM imaging modality (as opposed to, e.g., fluorescence microscopy), we reasoned that its inclusion as an augmentation may help decrease the reliance of the model on high-resolution features. We apply Gaussian blur to each volume with probability 0.5 after the affine transformation. If blurring is to be applied, we randomly select a blurring radius σ_{blur} from three options with equal chance (0.25, 0.5, or 0.75 in the final units of 14.3 nm), generate a normalized $3 \times 3 \times 3$ Gaussian 3D convolution kernel K , and apply the 3D convolution to the intensity tensor:

$$\sigma_{blur} \sim \text{Categorical}([0.25, 0.5, 0.75]) \quad (\text{Equation 6})$$

$$K = N(0, \sigma_{blur})$$

$$V_{out} = K * V_{in}$$

Mask-based transformations

Peripheral cutout

We encourage the model to attend to synapse active zone by disincentivizing it to use peripheral intensity data using three different transformations.

1. Corner cubic cutout: we subdivide the synaptic volume in thirds along each axis, randomly select one of the 26 cubes neighboring the central region, and blank it out with probability 0.5.
2. Random cubic cutout: we randomly select and blank out two cubic regions, each ranging in dimensions from $12 \times 12 \times 12$ voxels to $36 \times 36 \times 36$ voxels, with probability 0.25. We protect the synaptic cleft region from being selected and cropped.
3. Center crop: we select a cubic region around the synaptic cleft with linear dimensions ranging from 50% to 70% of the entire box, with a

maximum center displacement of 20%, and we blank out the exterior region with a probability of 0.25.

These transformations are applied after the affine transformation in the final $96 \times 96 \times 96$ voxels space.

Segmentation mask inflation

We disincentivize the model from exploiting potential imperfections of the segmentation masks by inflating (dilating) the segmentation masks with randomly chosen inflation radii. We implement the operation by convolving the binary masks with a cubic dilation structure as follows:

$$r_{inflation} \sim \text{Categorical}(1, \dots, 9) \quad (\text{Equation 7})$$

$$K = \text{Rect}(r_{inflation})$$

$$M_{out} = K *_{B} M_{in}$$

where $r_{inflation}$ is the inflation radius; $\text{Rect}(r_{inflation})$ is the cubic dilation structure, i.e., a 3D kernel with shape $(2r_{inflation} + 1)^3$ with 1 everywhere; M_{in} and M_{out} are the input and output binary segmentation masks; and $*_{B}$ is the binary convolution operator.

Data augmentation in batch mode

When augmenting a batch of images, we implement two modes of operation: (1) the coupled mode, where the same randomly composed augmentation transformation is applied to every image in the batch, and (2) the decoupled mode, where a different randomly composed transformation is generated and applied to each image. In theory, the coupled mode leads to both harder positive and negative examples: for any given image in a batch of n images, the same transformation is applied to $n - 1$ other images, while also the matching image (the second in the pair) shares the same augmentation transformation with $n - 1$ negative images. As a result, the contrastive loss cannot be trivially minimized by learning the signature of augmentation transformations and exploiting those signatures to rule out potential positives or negatives. In our preliminary experiments, we observed evidence supporting improved representations when training the model using coupled augmentations (e.g., a more structured UMAP embedding) in agreement with our theoretical reasoning. The results shown in this paper are all generated using models trained using coupled augmentations, a choice that deviates from the original SimCLR paper.⁵³ A systematic empirical study of the impact of different augmentation strategies on the quality of representations is a valuable future research direction.

Feature extraction mode

A critical part of our raw data preprocessing, namely bilinear upsampling of the axial resolution to match the in-plane resolution, and downsampling to the final $96 \times 96 \times 96$ resolution (which is an allowed input dimension for 3D-ResNet-18 CNN architecture), is done on the fly and as a part of our fast GPU-based data augmentation. Following successful model training, we wish to process each synapse with the trained CNN and extract representations from different layers (backbone, projection middle, projection head). To this end, we simply disable all intensity-based, mask-based, and affine transformations except for the anisotropic scaling component of the affine transformation step. In this mode, the augmentation module operates in a fully deterministic mode and simply acts as a fast data resolution transformer.

SynapseCLR loss function and training schedule

We closely follow the SimCLR methodology developed for natural images,⁵³ with two modifications: (1) we use a 3D-ResNet18 CNN as the backbone instead of 2D-ResNets; (2) we replace the SimCLR data augmentation transformations (relevant to 2D natural images) with the 3D EM-specific transformations discussed in the previous section. In this section, we provide a brief description of our adapted framework (SynapseCLR), including the neural network architectures, loss function, and training schedule. We implemented SynapseCLR using the PyTorch machine learning framework with support for distributed multiple-GPU training for scaling.

The SynapseCLR framework: Network architecture and loss function

Each iteration of SynapseCLR training starts with a minibatch of N randomly selected images, x_i for $i = 1, \dots, N$. Next, we generate two random views (augmentations) of each image by composing a series of randomly selected transformations, t_i for $i = 1, \dots, 2N$, from a predefined set of transformations. These transformations are applied to the images in the minibatch and the results are interleaved in even and odd pairs:

$$\tilde{x}_{2k-1} = t_{2k-1}(x_k) \quad (\text{Equation 8})$$

$$\tilde{x}_{2k} = t_{2k}(x_k),$$

for $k = 1, \dots, N$. The augmented images are first processed by the CNN backbone $f(\cdot)$, followed by the projection head $g(\cdot)$, in order to generate a final representation for contrasting and matching:

$$h_{2k-1} = f(\tilde{x}_{2k-1}), z_{2k-1} = g(h_{2k-1}) \quad (\text{Equation 9})$$

$$h_{2k} = f(\tilde{x}_{2k}), z_{2k} = g(h_{2k})$$

for $k = 1, \dots, N$. The role of the projection head is to further process the CNN features and embed them in a space amenable to contrasting and matching using one's similarity metric of choice (see below). The architecture of the 3D-ResNet18 and projection head we have used in our implementation is shown in Figure S1A. Our ResNet backbone is largely a 3D adaptation of the original 2D ResNet⁶⁵ with a minor modification of the network stem: given that our input images are already downsampled to $96 \times 96 \times 96$, with each voxel corresponding to ~ 14.3 nm in physical space, we reasoned that a maximum pooling operation commonly in the network stem would result in loss of high-frequency visual information and therefore bypassed it.

The SimCLR contrastive training objective is to encourage two views of the same image to be as similar as possible, i.e., $h_{2k-1} \approx h_{2k}$, while maintaining dissimilarity to both views of every other image in the minibatch. This is achieved by calculating and summing a normalized temperature-scaled cross entropy (NT-Xent) loss function for each image in the augmented minibatch:

$$L = \frac{1}{2N} \sum_{k=1}^N (l_{2k-1,2k} + l_{2k,2k-1}) \quad (\text{Equation 10})$$

where:

$$l_{ij} = -z_i^T z_j \left/ \tau + \log \sum_{k=1}^{2N} (1_{k \neq i} \exp(z_i^T z_k / \tau)) \right. \quad (\text{Equation 11})$$

Here, τ is the temperature hyperparameter that sets the typical range of repulsive and attractive forces in the representation space. It has been empirically shown that $\tau = 0.5$ is a robust choice independent of the learning rate and minibatch size.⁵³

Prior to model training, the projection head representations z_i are largely uninformative. As such, the cosine similarity values $z_i^T z_j$ appearing in l_{ij} are drawn from the same distribution for both positive and negative pairs. As a result, $l_{ij} \approx \log(2N - 1)$ independently of i and j and the initial value of the loss function is expected to be $L_{\text{init}} \approx \log(2N - 1)$. As model training proceeds, we expect the appearance of a separation in cosine similarity between positive and negative pairs, i.e., $z_i^T z_j - z_i^T z_k \approx \Delta > 0$, for i and j being two views of the same image and k being a different image. Consequently, we find $l_{2k-1,2k} \approx l_{2k,2k-1} \approx \log(2N - 1) - \Delta / \tau$. Therefore, a drop in the loss function L from its initial value by an amount δL implies the emergence of a separation $\Delta \approx \tau \delta L$ in the cosine similarity between positive and negative pairs.

Initialization and training schedule

We initialized the convolutional layers using Glorot-scaled Gaussian weights¹⁰⁰ truncated to $[-2, 2]$. Weights and biases of the linear layers were initialized to i.i.d. draws from $N(0, 0.01)$. The 3D-ResNet18 and projection head contain ~ 33.4 million and $\sim 329,000$ trainable parameters, respectively. We used the Adam optimizer with an initial learning rate of 2×10^{-4} with a

cosine decay to 2×10^{-5} during 100 training epochs. After 100 epochs, we reset the learning rate back to 2×10^{-4} and performed another 100 epochs of contrastive training. We used 4x NVIDIA A100 GPUs, each with 40 GB of HBM2 RAM. With 160 GB of GPU RAM, we could upload, augment, and perform a complete forward and backward pass on a minibatch size of 192 synapses (48 synapses/GPU). Training for 200 epochs took approximately 2 weeks. The NT-Xent loss function is shown in Figures S1B and is largely stationary after restart, suggesting convergence.

Annotations for verified synapses

For EM image chunks of verified synapses, we also annotated several properties that previous studies indicate may be related to synaptic function and thus potentially also of structure.

- Presynaptic cell type: assigned as excitatory or inhibitory from the soma valence table provided in <https://www.microns-explorer.org/phase1> (section "data and code availability").
- Postsynaptic cell type: also assigned as excitatory or inhibitory from the soma valence table.
- Synaptic cleft size: the number of voxels forming the synaptic cleft, as reported in the synapse table from the dataset.
- Distance from synaptic cleft to the soma of the presynaptic neuron: computed by finding (1) the node in the presynaptic neuron skeleton closest to the synaptic cleft centroid, and (2) the node of the presynaptic skeleton corresponding to the soma centroid. We then used the NetworkX¹⁰¹ package in Python to convert the presynaptic skeleton to a graph, and used the shortest_path_length algorithm to estimate the distance between these two locations.
- Distance from synaptic cleft to the soma of the postsynaptic neuron: computed using the same process as above with the postsynaptic neuron.
- Presence of presynaptic mitochondria: mitochondrial segmentations within 5 μm of annotated synapse centroids were manually proofread. If mitochondrial segmentations overlapped with the presynaptic segment in an annotated synapse image chunk, this value was set to 1; otherwise, it was set to 0.
- Presence of postsynaptic mitochondria: same as above, but mitochondria overlapping with the postsynaptic segment were used.
- Volume of presynaptic mitochondria: the total number of voxels of mitochondrial segments overlapping with the presynaptic segment that also were within the synapse image chunk (to reflect findings implicating mitochondrial proximity to a synapse as an important determinant of mitochondrial impact on synaptic function^{50,102}).
- Volume of postsynaptic mitochondria: same as above but for mitochondrial segments overlapping the postsynaptic segment.

GP regression for annotation imputation

In this section, we describe our methodology for probabilistic imputation of annotations for unlabeled synapses. Conceptually, we wish to use SynapseCLR representations as covariates (input space) and predict the annotations (target space) using available manually annotated synapses. Previous authors have used linear classifiers and regressors for this task.⁵³ Here, we propose to use GP models instead. GP has two key advantages compared with linear models: (1) linear models can be thought of as GPs with linear kernels, and using appropriate non-linear kernels may prove to be an effective way to improve the accuracy of predictions; (2) GP models are naturally endowed with a Bayesian posterior probability distribution, allowing us to additionally model the credibility of our predictions.

Annotation preprocessing and class balancing

To facilitate GP training, we log transformed and z scored all continuous annotations to compress and standardize their dynamic ranges. This transformation is invertible, allowing us to revert to the original scale after performing imputation. In light of the presence of significant sampling bias in our available annotations (e.g., there are more inhibitory synapses than excitatory synapses in the annotation set; see Figure 4D), it is crucial to perform class balancing on the GP training dataset. To this end, we randomly resampled an equal number of synapses from each binary class 5 times and trained separate GP models on

each rebalanced dataset. We observed reassuringly negligible differences in predictions and reported ensemble averages of the GP posterior predictions.

Cross-validation and accuracy metrics

For an unbiased assessment of the accuracy of the GP fits, we split the annotated dataset into 90% training and 10% testing sets, fit the GP on the training set, and reported accuracy figures on the testing set. We repeated this procedure five times with different random splits to estimate the confidence level of the reported accuracy figures. We report the prediction accuracy of continuous variables in terms of EV percentage over the test set defined as follows:

$$EV(x_{pred}, x_{test}) = 100 \times \left(1 - \frac{\sum_{i=1}^{N_{test}} (x_{test}^{(i)} - x_{pred}^{(i)})^2}{\sum_{i=1}^{N_{test}} (x_{test}^{(i)})^2} \right) \quad (\text{Equation 12})$$

where $x_{test}^{(i)}$ and $x_{pred}^{(i)}$ denote the expected and predicted values of a specific annotation for the i th synapse in the test set. For binary traits, we report the area under the receiver operating characteristic (precision-recall) curve (AUC), calculated using `sklearn.metrics.auc` function implemented in `scikit-learn`¹⁰³ (see section “software used” below).

GP fitting and hyperparameter optimization

We leveraged the GP functionality available in the Pyro probabilistic programming language.¹⁰⁴ In particular, we used the variational sparse GP (VSGP) flavor using a mean-field posterior for speed and scalability. Compared with ordinary GP regression, VSGP does not directly calculate the kernel over all of the data points. Instead, a specified number of representative points (called inducing points) are chosen and their location is optimized to obtain the best fit. Reducing the number of inducing points acts as a regularization and can be used to control overfitting to the training data.

We trained separate GP regressors and categorical classifiers for each continuous and discrete annotation. In principle, better fits might be achieved via multi-task training (co-kriging); however, we did not attempt that here. We trained separate GP models on 512-dimensional CNN backbone, 512-dimensional projection middle, and 128-dimensional projection head features as covariates. In each case, we initialized the VSGP inducing points using k-means. We used the Adam optimizer with an initial learning rate of 1×10^{-3} with cosine annealing to 0 during 20,000 training epochs. We considered performing PCA dimensionality reduction on the covariates before fitting; however, we found inferior performance in every case (see below).

We used the previously described cross-validation strategy and accuracy metrics to optimize (1) the number of VSGP inducing points, and (2) the choice of GP kernel. A summary of the experiments is shown in Table S1, pointing to the following general conclusions:

1. Using the 512-dimensional CNN backbone features as covariates provided the highest prediction accuracy for all of the annotations (see the first group of rows in Table S1).
2. PCA dimensionality reduction of covariates, regardless of the number of kept PCs, degraded the accuracy of predictions (see the second group of rows in Table S1).
3. We considered the following kernels:
 - a. Linear kernel with white noise:

$$K_{Linear}(z, z') = \sigma^2 (z \cdot z') + \theta + \sigma_{white}^2 \delta(z, z')$$

- b. Laplace kernel with white noise:

$$K_{Laplace}(z, z') = \sigma^2 e^{-|z - z'|/l} + \sigma_{white}^2 \delta(z, z')$$

- c. Radial basis function (RBF) kernel with white noise:

$$K_{RBF}(z, z') = \sigma^2 e^{-|z - z'|^2 / (2l^2)} + \sigma_{white}^2 \delta(z, z')$$

The best fits were obtained using the RBF kernel with white noise and with a fine-tuned number of inducing points. The latter effectively acts as a regularization by controlling the model complexity, allowing us to prevent overfitting, which is made possible using the more flexible class of functions reproduced by the RBF kernel (see the third group of rows in Table S1).

Finally, Figure 3A shows the accuracy of best GP fits obtained by following the same comprehensive hyperparameter optimization strategy using features extracted from a random CNN baseline and MedicalNet, a general-purpose pretrained 3D-ResNet18 CNN. We found that SynapseCLR features lead to significantly higher GP imputation accuracy figures.

Ensembling and consensus-based cell typing

We showed that SynapseCLR representation learning followed by fitting GP classifiers can reliably predict pre- and postsynaptic cell types for individual synapses (0.975 and 0.940 AUC, respectively; see Figure 3A). As mentioned in the main text, we may further exploit the fact that a typical synapse-forming neurite in our dataset contains tens of synapses. This figure reaches thousands of synapses for unfragmented neurites. A highly effective way to boost the accuracy of several weak classifiers is by ensembling; i.e., by taking the consensus of all weak classifiers. In the present context, single-synapse cell type predictions can be thought of as weak classifications. By appropriately chaining weak predictions over synapses sharing the same neurite, the neurite-level cell type prediction error decreases rapidly and reaches practically noiseless limits. The consensus cell type of each neurite can then be propagated back to the participating synapses (see Figure 4A). We carry out this procedure in two steps.

Step 1 (neurite cell typing)

We use the 3D segmentation and skeletonization information available to us to compile a list of synapses formed onto each neurite. We proceed to calculate a consensus cell type probability for each neurite. To this end, we use a minimal Bayesian model in which the identity of a given neurite determines the pre- and postsynaptic cell type probabilities p_n of individual synapses formed onto it:

$$p_n | \alpha, \beta \sim \text{Beta}(\alpha, \beta) \quad (\text{Equation 13})$$

where $n = 1, \dots, N_{syn}$ enumerate the synapses formed onto the neurite. We assume $p_n = 0$ and $p_n = 1$ for excitatory and inhibitory cell types, respectively. We assume p_n are i.i.d. conditioned on α and β , the concentration parameters of a Beta distribution that determines the cell type of the neurite. The maximum likelihood procedure for estimating α and β yields the following set of coupled equations:

$$-N_{syn} [\psi(\alpha) - \psi(\alpha + \beta)] + \sum_{n=1}^{N_{syn}} \log p_n = 0 \quad (\text{Equation 14})$$

$$-N_{syn} [\psi(\beta) - \psi(\alpha + \beta)] + \sum_{n=1}^{N_{syn}} \log(1 - p_n) = 0$$

where $\psi(\cdot)$ is the digamma function. We can easily establish the following identity for the expected log odds of the excitatory class, $\log\left(\frac{p}{1-p}\right)$ using the basic properties of Beta distribution:

$$\begin{aligned} LO_{consensus} &= E \left[\log \left(\frac{p}{1-p} \right) \right] = \psi(\alpha) - \psi(\beta) \\ &= \frac{1}{N} \left[\sum_{n=1}^{N_{syn}} \log p_n - \sum_{n=1}^{N_{syn}} \log(1 - p_n) \right] \end{aligned} \quad (\text{Equation 15})$$

where the last equality results from maximum likelihood equations given earlier. According to this simple model, the consensus procedure amounts to summing up the log probabilities of excitatory class predictions and

subtracting the log probabilities of inhibitory class predictions. We make a consensus cell type call for a neurite depending on the expected log odds: inhibitory if $LO_{consensus} > 0$, and excitatory if $LO_{consensus} < 0$.

Step 2 (synapse consensus cell typing)

We simply assign the consensus cell type of a neurite to the participating synapses according to the polarity of the synapse: if the neurite is the presynaptic partner of synapse, the neurite consensus cell type is thought of as the consensus presynaptic cell type of the synapse, and similarly for postsynaptic neurites.

Software used

CloudVolume⁹⁶

<https://github.com/seung-lab/cloud-volume>.

PyTorch⁹⁷

<https://github.com/pytorch/pytorch>.

cc3d⁹⁹

<https://github.com/seung-lab/connected-components-3d/>.

NetworkX¹⁰¹

<https://networkx.org/>.

NumPy¹⁰⁵

<https://numpy.org/citing-numpy/>.

SciPy¹⁰⁶

<https://scipy.org>.

Pandas¹⁰⁷

<https://github.com/pandas-dev/pandas>.

Matplotlib¹⁰⁸

<https://github.com/matplotlib/matplotlib>.

scikit-learn¹⁰³

<https://github.com/scikit-learn/scikit-learn>.

cuML⁷⁷

<https://github.com/rapidsai/cuml>.

Software referenced

MicronsBinder

<https://github.com/AllenInstitute/MicronsBinder>.

Original software

SynapseCLR and SynapseAugmenter Python packages.

Jupyter notebooks for recreating all of the figures in the paper.

Code for visualizing image chunks and segmentation layers in Python.

Code for assigning consensus cell types to neurites and synapses.

Code for imputation and prediction cross-validation of synapse annotations using SynapseCLR representations (see section “[data and code availability](#)” above).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.patter.2023.100693>.

ACKNOWLEDGMENTS

The authors thank Stephen Fleming, Luca D'Alessio, Nicholas Barkas, Anthony Philippakis, and William Silversmith for insightful discussions at various stages of this project. A.W. acknowledges financial support from NIH NIMH BRAIN Initiative award number RF1MH123400, IARPA contract number D16PC0005, and NIH NIMH award number T32MH065214. M.B. acknowledges financial support from Data Sciences Platform (DSP), Broad Institute. M.B. was additionally supported by a BroadIgnite award.

AUTHOR CONTRIBUTIONS

A.W. and M.B. contributed equally to the conceptualization of this work, methodology design, formal analysis and validation, and manuscript writing. A.W. executed data and annotation curation with near-equal involvement from M.B. M.B. executed software development with near-equal involvement from A.W.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

Received: July 11, 2022

Revised: November 23, 2022

Accepted: January 25, 2023

Published: March 7, 2023

REFERENCES

- Abbott, L.F., Bock, D.D., Callaway, E.M., Denk, W., Dulac, C., Fairhall, A.L., Fiete, I., Harris, K.M., Helmstaedter, M., Jain, V., et al. (2020). The mind of a mouse. *Cell* 182, 1372–1376. <https://doi.org/10.1016/j.cell.2020.08.010>.
- Turner, N.L., Macrina, T., Bae, J.A., Yang, R., Wilson, A.M., Schneider-Mizell, C., Lee, K., Lu, R., Wu, J., Bodor, A.L., et al. (2022). Reconstruction of neocortex: organelles, compartments, cells, circuits, and activity. *Cell* 185, 1082–1100.e24. <https://doi.org/10.1016/j.cell.2022.01.023>.
- Shapson-Coe, A., Januszewski, M., Berger, D.R., and Pope, A. (2021). A connectomic study of a petascale fragment of human cerebral cortex. Preprint at bioRxiv. <https://doi.org/10.1101/2021.05.29.446289>.
- Kasthuri, N., Hayworth, K.J., Berger, D.R., Schalek, R.L., Conchello, J.A., Knowles-Barley, S., Lee, D., Vázquez-Reina, A., Kaynig, V., Jones, T.R., et al. (2015). Saturated reconstruction of a volume of neocortex. *Cell* 162, 648–661. <https://doi.org/10.1016/j.cell.2015.06.054>.
- Briggman, K.L., Helmstaedter, M., and Denk, W. (2011). Wiring specificity in the direction-selectivity circuit of the retina. *Nature* 471, 183–188. <https://doi.org/10.1038/nature09818>.
- Bae, J.A., Mu, S., Kim, J.S., Turner, N.L., Tartavull, I., Kemnitz, N., Jordan, C.S., Norton, A.D., Silversmith, W.M., Prentki, R., et al. (2018). Digital museum of retinal ganglion cells with dense anatomy and physiology. *Cell* 173, 1293–1306.e19. <https://doi.org/10.1016/j.cell.2018.04.040>.
- Kim, J.S., Greene, M.J., Zlateski, A., Lee, K., Richardson, M., Turaga, S.C., Purcaro, M., Balkam, M., Robinson, A., Behabadi, B.F., et al. (2014). Space-time wiring specificity supports direction selectivity in the retina. *Nature* 509, 331–336. <https://doi.org/10.1038/nature13240>.
- Helmstaedter, M., Briggman, K.L., Turaga, S.C., Jain, V., Seung, H.S., and Denk, W. (2013). Connectomic reconstruction of the inner plexiform layer in the mouse retina. *Nature* 500, 168–174. <https://doi.org/10.1038/nature12346>.
- Behrens, C., Schubert, T., Haverkamp, S., Euler, T., and Berens, P. (2016). Connectivity map of bipolar cells and photoreceptors in the mouse retina. *Elife* 5, e20041. <https://doi.org/10.7554/eLife.20041>.
- Bock, D.D., Lee, W.-C.A., Kerlin, A.M., Andermann, M.L., Hood, G., Wetzel, A.W., Yurgenson, S., Soucy, E.R., Kim, H.S., and Reid, R.C. (2011). Network anatomy and in vivo physiology of visual cortical neurons. *Nature* 471, 177–182. <https://doi.org/10.1038/nature09802>.
- Zheng, Z., Lauritzen, J.S., Perlman, E., Robinson, C.G., Nichols, M., Milkie, D., Torrens, O., Price, J., Fisher, C.B., Sharifi, N., et al. (2018). A complete electron microscopy volume of the brain of adult *Drosophila melanogaster*. *Cell* 174, 730–743.e22. <https://doi.org/10.1016/j.cell.2018.06.019>.
- Karimi, A., Odenthal, J., Drawitsch, F., Boergens, K.M., and Helmstaedter, M. (2020). Cell-type specific innervation of cortical pyramidal cells at their apical dendrites. *Elife* 9, e46876. <https://doi.org/10.7554/eLife.46876>.
- Kornfeld, J., Benezra, S.E., Narayanan, R.T., Svava, F., Egger, R., Oberlaender, M., Denk, W., and Long, M.A. (2017). EM connectomics reveals axonal target variation in a sequence-generating network. *Elife* 6, e24364. <https://doi.org/10.7554/eLife.24364>.

14. Lee, W.-C.A., Bonin, V., Reed, M., Graham, B.J., Hood, G., Glatfelter, K., and Reid, R.C. (2016). Anatomy and function of an excitatory network in the visual cortex. *Nature* 532, 370–374. <https://doi.org/10.1038/nature17192>.
15. Mishchenko, Y., Hu, T., Spacek, J., Mendenhall, J., Harris, K.M., and Chklovskii, D.B. (2010). Ultrastructural analysis of hippocampal neuropil from the connectomics perspective. *Neuron* 67, 1009–1020. <https://doi.org/10.1016/j.neuron.2010.08.014>.
16. Morgan, J.L., Berger, D.R., Wetzel, A.W., and Lichtman, J.W. (2016). The Fuzzy logic of network connectivity in mouse visual thalamus. *Cell* 165, 192–206. <https://doi.org/10.1016/j.cell.2016.02.033>.
17. Motta, A., Berning, M., Boergens, K.M., Staffler, B., Beining, M., Lombar, S., Hennig, P., Wissler, H., and Helmstaedter, M. (2019). Dense connectomic reconstruction in layer 4 of the somatosensory cortex. *Science* 366, eaay3134. <https://doi.org/10.1126/science.aay3134>.
18. Schmidt, H., Gour, A., Straehle, J., Boergens, K.M., Brecht, M., and Helmstaedter, M. (2017). Axonal synapse sorting in medial entorhinal cortex. *Nature* 549, 469–475. <https://doi.org/10.1038/nature24005>.
19. Zonouzi, M., Berger, D., Jokhi, V., Kedaigle, A., Lichtman, J., and Ariotta, P. (2019). Individual Oligodendrocytes show bias for inhibitory axons in the neocortex. *Cell Rep.* 27, 2799–2808.e3. <https://doi.org/10.1016/j.celrep.2019.05.018>.
20. Shan Xu, C., Januszewski, M., Lu, Z., Takemura, S.-Y., Hayworth, K.J., Huang, G., Shinomiya, K., Maitin-Shepard, J., Ackerman, D., Berg, S., et al. (2020). A connectome of the adult *Drosophila* central brain. Preprint at bioRxiv. <https://doi.org/10.1101/2020.01.21.911859>.
21. Takemura, S.-Y., Bharioke, A., Lu, Z., Nern, A., Vitaladevuni, S., Rivlin, P.K., Katz, W.T., Olbris, D.J., Plaza, S.M., Winston, P., et al. (2013). A visual motion detection circuit suggested by *Drosophila* connectomics. *Nature* 500, 175–181. <https://doi.org/10.1038/nature12450>.
22. Takemura, S.-Y., Aso, Y., Hige, T., Wong, A., Lu, Z., Xu, C.S., Rivlin, P.K., Hess, H., Zhao, T., Parag, T., et al. (2017). A connectome of a learning and memory center in the adult *Drosophila* brain. *Elife* 6, e26975. <https://doi.org/10.7554/eLife.26975>.
23. Tapia, J.C., Wylie, J.D., Kasthuri, N., Hayworth, K.J., Schalek, R., Berger, D.R., Guatimosim, C., Seung, H.S., and Lichtman, J.W. (2012). Pervasive synaptic branch removal in the mammalian neuromuscular system at birth. *Neuron* 74, 816–829. <https://doi.org/10.1016/j.neuron.2012.04.017>.
24. Vishwanathan, A., Daie, K., Ramirez, A.D., Lichtman, J.W., Aksay, E.R.F., and Seung, H.S. (2017). Electron microscopic reconstruction of functionally identified cells in a neural integrator. *Curr. Biol.* 27, 2137–2147.e3. <https://doi.org/10.1016/j.cub.2017.06.028>.
25. Wanner, A.A., and Friedrich, R.W. (2020). Whitening of odor representations by the wiring diagram of the olfactory bulb. *Nat. Neurosci.* 23, 433–442. <https://doi.org/10.1038/s41593-019-0576-z>.
26. Helmstaedter, M., Briggman, K.L., and Denk, W. (2011). High-accuracy neurite reconstruction for high-throughput neuroanatomy. *Nat. Neurosci.* 14, 1081–1088. <https://doi.org/10.1038/nn.2868>.
27. Hoggarth, A., McLaughlin, A.J., Ronellenfitch, K., Trenholm, S., Vasandani, R., Sethuramanujam, S., Schwab, D., Briggman, K.L., and Awatramani, G.B. (2015). Specific wiring of distinct amacrine cells in the directionally selective retinal circuit permits independent coding of direction and size. *Neuron* 86, 276–291. <https://doi.org/10.1016/j.neuron.2015.02.035>.
28. Ding, H., Smith, R.G., Poleg-Polsky, A., Diamond, J.S., and Briggman, K.L. (2016). Species-specific wiring for direction selectivity in the mammalian retina. *Nature* 535, 105–110. <https://doi.org/10.1038/nature18609>.
29. Dorkenwald, S., McKellar, C., Macrina, T., Kemnitz, N., Lee, K., Lu, R., Wu, J., Popovych, S., Mitchell, E., Nehoran, B., et al. (2020). FlyWire: online community for whole-brain connectomics. Preprint at bioRxiv. <https://doi.org/10.1101/2020.08.30.274225>.
30. Wilson, A.M., Schalek, R., Suissa-Peleg, A., Jones, T.R., Knowles-Barley, S., Pfister, H., and Lichtman, J.W. (2019). Developmental rewiring between cerebellar climbing fibers and purkinje cells begins with positive feedback synapse addition. *Cell Rep.* 29, 2849–2861.e6. <https://doi.org/10.1016/j.celrep.2019.10.081>.
31. Bloss, E.B., Cembrowski, M.S., Karsh, B., Colonell, J., Fetter, R.D., and Spruston, N. (2018). Single excitatory axons form clustered synapses onto CA1 pyramidal cell dendrites. *Nat. Neurosci.* 21, 353–363. <https://doi.org/10.1038/s41593-018-0084-6>.
32. Gour, A., Boergens, K.M., Heike, N., Hua, Y., Laserstein, P., Song, K., and Helmstaedter, M. (2021). Postnatal connectomic development of inhibition in mouse barrel cortex. *Science* 371, eabb4534. <https://doi.org/10.1126/science.abb4534>.
33. Witvliet, D., Mulcahy, B., Mitchell, J.K., Meirovitch, Y., Berger, D.R., Wu, Y., Liu, Y., Koh, W.X., Parvathala, R., Holmyard, D., et al. (2021). Connectomes across development reveal principles of brain maturation. *Nature* 596, 257–261. <https://doi.org/10.1038/s41586-021-03778-8>.
34. Bartol, T.M., Bromer, C., Kinney, J., Chirillo, M.A., Bourne, J.N., Harris, K.M., and Sejnowski, T.J. (2015). Nanoconnectomic upper bound on the variability of synaptic plasticity. *Elife* 4, e10778. <https://doi.org/10.7554/eLife.10778>.
35. Karube, F., Kubota, Y., and Kawaguchi, Y. (2004). Axon branching and synaptic bouton phenotypes in GABAergic nonpyramidal cell subtypes. *J. Neurosci.* 24, 2853–2865. <https://doi.org/10.1523/JNEUROSCI.4814-03.2004>.
36. Wu, Y., Whiteus, C., Xu, C.S., Hayworth, K.J., Weinberg, R.J., Hess, H.F., and De Camilli, P. (2017). Contacts between the endoplasmic reticulum and other membranes in neurons. *Proc. Natl. Acad. Sci. USA* 114, E4859–E4867. <https://doi.org/10.1073/pnas.1701078114>.
37. Zhu, Y., Uytiepo, M., Bushong, E., Haberl, M., Beutler, E., Scheiwe, F., Zhang, W., Chang, L., Luu, D., Chui, B., et al. (2021). Nanoscale 3D EM reconstructions reveal intrinsic mechanisms of structural diversity of chemical synapses. *Cell Rep.* 35, 108953. <https://doi.org/10.1016/j.celrep.2021.108953>.
38. Buchanan, J., Elabbady, L., Collman, F., Jorstad, N.L., Bakken, T.E., Ott, C., Glatzer, J., Bleckert, A.A., Bodor, A.L., Brittain, D., et al. (2022). Oligodendrocyte precursor cells ingest axons in the mouse neocortex. *Proc. Natl. Acad. Sci. USA* 119, e2202580119. <https://doi.org/10.1073/pnas.2202580119>.
39. Dorkenwald, S., Turner, N.L., Macrina, T., Lee, K., Lu, R., Wu, J., Bodor, A.L., Bleckert, A.A., Brittain, D., Kemnitz, N., et al. (2019). Binary and analog variation of synapses between cortical pyramidal neurons. Preprint at bioRxiv. <https://doi.org/10.1101/2019.12.29.890319>.
40. Lichtman, J.W., and Denk, W. (2011). The big and the small: challenges of imaging the brain's circuits. *Science* 334, 618–623. <https://doi.org/10.1126/science.1209168>.
41. MICrONS Consortium, Alexander Bae, J., Baptiste, M., Bodor, A.L., Brittain, D., Buchanan, J., Bumbarger, D.J., Castro, M.A., Celli, B., Cobos, E., et al. (2021). Functional connectomics spanning multiple areas of mouse visual cortex. Preprint at bioRxiv. <https://doi.org/10.1101/2021.07.28.454025>.
42. Conrad, R., and Narayan, K. (2022). Instance segmentation of mitochondria in electron microscopy images with a generalist deep learning model. Preprint at bioRxiv. <https://doi.org/10.1101/2022.03.17.484806>.
43. Lee, K., Turner, N., Macrina, T., Wu, J., Lu, R., and Seung, H.S. (2019). Convolutional nets for reconstructing neural circuits from brain images acquired by serial section electron microscopy. *Curr. Opin. Neurobiol.* 55, 188–198. <https://doi.org/10.1016/j.conb.2019.04.001>.
44. Huang, G.B., Scheffer, L.K., and Plaza, S.M. (2018). Fully-automatic synapse prediction and validation on a large data set. *Front. Neural Circuits* 12, 87. <https://doi.org/10.3389/fncir.2018.00087>.
45. Luo, J., Hong, B., Jiang, Y., Li, L., Xie, Q., and Han, H. (2019). Automatic classification for the type of multiple synapse based on deep learning. *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.* 2019, 40–43. <https://doi.org/10.1109/EMBC.2019.8856509>.

46. Bailey, C.H., and Chen, M. (1983). Morphological basis of long-term habituation and sensitization in Aplysia. *Science* 220, 91–93. <https://doi.org/10.1126/science.6828885>.
47. Zhang, A., Shailja, S., Borba, C., Miao, Y., Goebel, M., Ruschel, R., Ryan, K., Smith, W., and Manjunath, B.S. (2021). Automatic detection and neurotransmitter prediction of synapses in electron microscopy. Preprint at bioRxiv. <https://doi.org/10.1101/2021.11.02.467022>.
48. Eckstein, N., Bates, A.S., Du, M., Hartenstein, V., Jefferis, G.S.X., and Funke, J. (2020). Neurotransmitter classification from electron microscopy images at synaptic Sites in Drosophila. Preprint at bioRxiv. <https://doi.org/10.1101/2020.06.12.148775>.
49. Dorkenwald, S., Schubert, P.J., Killinger, M.F., Urban, G., Mikula, S., Svava, F., and Kornfeld, J. (2017). Automated synaptic connectivity inference for volume electron microscopy. *Nat. Methods* 14, 435–442. <https://doi.org/10.1038/nmeth.4206>.
50. Rangaraju, V., Lewis, T.L., Jr., Hirabayashi, Y., Bergami, M., Motori, E., Cartoni, R., Kwon, S.-K., and Courchet, J. (2019). Pleiotropic mitochondria: the influence of mitochondria on neuronal development and disease. *J. Neurosci.* 39, 8200–8208. <https://doi.org/10.1523/JNEUROSCI.1157-19.2019>.
51. Singer, W., Sejnowski, T.J., and Rakic, P. (2019). *The Neocortex* (MIT Press).
52. Harris, K.M. (2020). Structural LTP: from synaptogenesis to regulated synapse enlargement and clustering. *Curr. Opin. Neurobiol.* 63, 189–197. <https://doi.org/10.1016/j.conb.2020.04.009>.
53. Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning* (PMLR), pp. 1597–1607.
54. Liu, X., Zhang, F., Hou, Z., Wang, Z., Mian, L., Zhang, J., and Tang, J. (2020). Self-supervised learning: generative or contrastive. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2006.08218>.
55. Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2002.05709>.
56. Chen, T., Kornblith, S., Swersky, K., Norouzi, M., and Hinton, G. (2020). Big self-supervised models are strong semi-supervised learners. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2006.10029>.
57. He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. (2019). Momentum contrast for unsupervised visual representation learning. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1911.05722>.
58. Grill, J.-B., Strub, F., Althé, F., Tallec, C., Richemond, P.H., Buchatskaya, E., Doersch, C., Pires, B.A., Guo, Z.D., Azar, M.G., et al. (2020). Bootstrap your own latent: a new approach to self-supervised learning. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2006.07733>.
59. Zbontar, J., Jing, L., Misra, I., LeCun, Y., and Deny, S. (2021). Barlow Twins: self-supervised learning via redundancy reduction. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2103.03230>.
60. Su, J.-C., Maji, S., and Hariharan, B. (2019). When does self-supervision improve few-shot learning?. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1910.03560>.
61. Li, J., Zhou, P., Xiong, C., and Hoi, S.C.H. (2020). Prototypical contrastive learning of unsupervised representations. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2005.04966>.
62. Fukushima, K. (1980). Neocognitron: a self organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biol. Cybern.* 36, 193–202. <https://doi.org/10.1007/BF00344251>.
63. Feichtenhofer, C., Pinz, A., and Wildes, R.P. (2016). Spatiotemporal Residual Networks for video action recognition. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1611.02155>.
64. Hara, K., Kataoka, H., and Satoh, Y. (2017). Learning spatio-temporal features with 3D residual networks for action recognition. In *2017 IEEE International Conference on Computer Vision Workshops (ICCVW) (IEEE)*. <https://doi.org/10.1109/iccvw.2017.373>.
65. He, K., Zhang, X., Ren, S., and Sun, J. (2015). Deep residual learning for image recognition. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1512.03385>.
66. Wu, Y., Dissing-Olesen, L., MacVicar, B.A., and Stevens, B. (2015). Microglia: dynamic Mediators of synapse development and plasticity. *Trends Immunol.* 36, 605–613. <https://doi.org/10.1016/j.it.2015.08.008>.
67. Stepanyants, A., and Chklovskii, D.B. (2005). Neurogeometry and potential synaptic connectivity. *Trends Neurosci.* 28, 387–394. <https://doi.org/10.1016/j.tins.2005.05.006>.
68. McInnes, L., Healy, J., and Melville, J. (2018). UMAP: uniform manifold approximation and projection for dimension reduction. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1802.03426>.
69. Traag, V.A., Waltman, L., and van Eck, N.J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Sci. Rep.* 9, 5233. <https://doi.org/10.1038/s41598-019-41695-z>.
70. Chen, S., Ma, K., and Zheng, Y. (2019). Med3D: transfer learning for 3D Medical image analysis. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1904.00625>.
71. Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. (2009). ImageNet: a large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255. <https://doi.org/10.1109/CVPR.2009.5206848>.
72. Bychkov, D., Linder, N., Turkki, R., Nordling, S., Kovanen, P.E., Verrill, C., Wallander, M., Lundin, M., Haglund, C., and Lundin, J. (2018). Deep learning based tissue analysis predicts outcome in colorectal cancer. *Sci. Rep.* 8, 3395. <https://doi.org/10.1038/s41598-018-21758-3>.
73. Caicedo, J.C., McQuin, C., Goodman, A., Singh, S., and Carpenter, A.E. (2018). Weakly supervised learning of single-cell feature embeddings. *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* 2018, 9309–9318. <https://doi.org/10.1109/CVPR.2018.00970>.
74. Ester, M., Kriegel, H.P., Sander, J., and Xu, X. (1996). *A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise* (KDD).
75. Schneider-Mizell, C.M., Bodor, A.L., Collman, F., Brittain, D., Bleckert, A., Dorkenwald, S., Turner, N.L., Macrina, T., Lee, K., Lu, R., et al. (2021). Structure and function of axo-axonic inhibition. *Elife* 10, e73783. <https://doi.org/10.7554/eLife.73783>.
76. Kubota, Y., Karube, F., Nomura, M., and Kawaguchi, Y. (2016). The diversity of cortical inhibitory synapses. *Front. Neural Circuits* 10, 27. <https://doi.org/10.3389/fncir.2016.00027>.
77. Raschka, S., Patterson, J., and Nolet, C. (2020). Machine learning in Python: main developments and Technology Trends in data science, machine learning, and artificial intelligence. *Information* 11, 193. <https://doi.org/10.3390/info11040193>.
78. Bopp, R., Maçarico da Costa, N., Kampa, B.M., Martin, K.A.C., and Roth, M.M. (2014). Pyramidal cells make specific connections onto smooth (GABAergic) neurons in mouse visual cortex. *PLoS Biol.* 12, e1001932. <https://doi.org/10.1371/journal.pbio.1001932>.
79. Yin, W., Brittain, D., Borseth, J., Scott, M.E., Williams, D., Perkins, J., Own, C.S., Murfitt, M., Torres, R.M., Kapner, D., et al. (2020). A petascale automated imaging pipeline for mapping neuronal circuits with high-throughput transmission electron microscopy. *Nat. Commun.* 11, 4949. <https://doi.org/10.1038/s41467-020-18659-3>.
80. Wang, Y., Yao, Q., Kwok, J.T., and Ni, L.M. (2020). Generalizing from a few examples: a survey on few-shot learning. *ACM Comput. Surv.* 53, 1–34. <https://doi.org/10.1145/3386252>.
81. Jackman, S.L., and Regehr, W.G. (2017). The mechanisms and functions of synaptic facilitation. *Neuron* 94, 447–464. <https://doi.org/10.1016/j.neuron.2017.02.047>.
82. Nicoll, R.A. (2017). A brief history of long-term potentiation. *Neuron* 93, 281–290. <https://doi.org/10.1016/j.neuron.2016.12.015>.

83. Vaccaro, V., Devine, M.J., Higgs, N.F., and Kittler, J.T. (2017). Miro1-dependent mitochondrial positioning drives the rescaling of presynaptic Ca²⁺ signals during homeostatic plasticity. *EMBO Rep.* 18, 231–240. <https://doi.org/10.15252/embr.201642710>.
84. Siddoway, B., Hou, H., and Xia, H. (2014). Molecular mechanisms of homeostatic synaptic downscaling. *Neuropharmacology* 78, 38–44. <https://doi.org/10.1016/j.neuropharm.2013.07.009>.
85. Tononi, G., and Cirelli, C. (2020). Sleep and synaptic down-selection. *Eur. J. Neurosci.* 51, 413–421. <https://doi.org/10.1111/ejn.14335>.
86. Kusick, G.F., Ogunmowo, T.H., and Watanabe, S. (2022). Transient docking of synaptic vesicles: implications and mechanisms. *Curr. Opin. Neurobiol.* 74, 102535. <https://doi.org/10.1016/j.conb.2022.102535>.
87. Higgins, I., Amos, D., Pfau, D., Racaniere, S., Matthey, L., Rezende, D., and Lerchner, A. (2018). Towards a definition of disentangled representations. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1812.02230>.
88. Zimmermann, R.S., Sharma, Y., Schneider, S., Bethge, M., and Brendel, W. (2021). Contrastive learning inverts the data generating process. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2102.08850>.
89. Pereda, A.E. (2014). Electrical synapses and their functional interactions with chemical synapses. *Nat. Rev. Neurosci.* 15, 250–263. <https://doi.org/10.1038/nrn3708>.
90. Carter, S.D., Hampton, C.M., Langlois, R., Melero, R., Farino, Z.J., Calderon, M.J., Li, W., Wallace, C.T., Tran, N.H., Grassucci, R.A., et al. (2020). Ribosome-associated vesicles: a dynamic subcompartment of the endoplasmic reticulum in secretory cells. *Sci. Adv.* 6, eaay9572. <https://doi.org/10.1126/sciadv.aay9572>.
91. Wakui, T., Negishi, M., Murakami, Y., Tominaga, S., Shiraishi, Y., Carpenter, A.E., Singh, S., and Segawa, H. (2022). Predicting gene expression from cell morphology in human induced pluripotent stem cells. Preprint at bioRxiv. <https://doi.org/10.1101/2022.04.19.488786>.
92. Schmauch, B., Romagnoni, A., Pronier, E., Saillard, C., Maillé, P., Calderaro, J., Kamoun, A., Sefta, M., Toldo, S., Zaslavskiy, M., et al. (2020). A deep learning model to predict RNA-Seq expression of tumours from whole slide images. *Nat. Commun.* 11, 3877. <https://doi.org/10.1038/s41467-020-17678-4>.
93. Ash, J.T., Darnell, G., Munro, D., and Engelhardt, B.E. (2021). Joint analysis of expression levels and histological images identifies genes associated with tissue morphology. *Nat. Commun.* 12, 1609. <https://doi.org/10.1038/s41467-021-21727-x>.
94. Rohban, M.H., Singh, S., Wu, X., Berthet, J.B., Bray, M.-A., Shrestha, Y., Varelas, X., Boehm, J.S., and Carpenter, A.E. (2017). Systematic morphological profiling of human gene and allele function via cell painting. *Elife* 6, e24060. <https://doi.org/10.7554/eLife.24060>.
95. Turner, N., Lee, K., Lu, R., Wu, J., Ih, D., and Sebastian Seung, H. (2019). Synaptic partner assignment using attentional voxel association networks. Preprint at arXiv. <https://doi.org/10.1109/ISBI45749.2020.9098489>.
96. Silversmith, W., Collman, F., Kemnitz, N., Wu, J., Castro, M., Falk, B., Roat, C., Macrina, T., Perlman, E., Mu, S., et al. (2021). seung-lab/cloud-volume: Zenodo Release 1. <https://doi.org/10.5281/zenodo.5671443>.
97. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimeshe, N., Antiga, L., et al. (2019). PyTorch: an imperative style, high-performance deep learning library. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1912.01703>.
98. Jaderberg, M., Simonyan, K., Zisserman, A., and Kavukcuoglu, K. (2015). Spatial transformer networks. *Adv. Neural Inf. Process. Syst.* 28.
99. Silversmith, W. (2021). seung-lab/connected-components-3d: Zenodo Release V1. <https://doi.org/10.5281/zenodo.5535251>.
100. Glorot, X., and Bengio, Y.. Understanding the Difficulty of Training Deep Feedforward Neural Networks. <https://proceedings.mlr.press/v9/glorot10a/glorot10a.pdf>.
101. Hagberg, A., Schult, D., and Swart, P. (2008). Exploring network structure, dynamics, and function using NetworkX. *Proc. 7th Python Sci. Conf. 2008*, 11–15.
102. Rangaraju, V., Lauterbach, M., and Schuman, E.M. (2019). Spatially stable mitochondrial compartments fuel local translation during plasticity. *Cell* 176, 73–84.e15. <https://doi.org/10.1016/j.cell.2018.12.013>.
103. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: machine learning in python. *J. Mach. Learn. Res.* 12, 2825–2830.
104. Bingham, E., Chen, J.P., Jankowiak, M., Obermeyer, F., Pradhan, N., Karaletos, T., Singh, R., Szerlip, P., Horsfall, P., and Goodman, N.D. (2018). Pyro: deep Universal probabilistic programming. Preprint at arXiv. <https://doi.org/10.48550/arXiv.1810.09538>.
105. Harris, C.R., Millman, K.J., van der Walt, S.J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N.J., et al. (2020). Array programming with NumPy. *Nature* 585, 357–362. <https://doi.org/10.1038/s41586-020-2649-2>.
106. Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., et al. (2020). SciPy 1.0: fundamental algorithms for scientific computing in python. *Nat. Methods* 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>.
107. The Pandas Development Team (2022). Pandas-Dev/Pandas: Pandas. <https://doi.org/10.5281/zenodo.7223478>.
108. Hunter, J.D. (2007). Matplotlib: a 2D graphics environment. *Comput. Sci. Eng.* 9, 90–95. <https://doi.org/10.1109/MCSE.2007.55>.